

2025 Chipathon 5th Weekly Project Proposals 1/2

IEEE Solid-State Circuits Society
Technical Committee on the Open Source Ecosystem (TC-OSE)
July 11, 2025



2025 Chipathon: Blocks & Bots: An Open Chip Playground augmented with LLMs

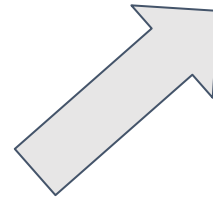


- ▶ **Phase 1** : Setup and Introduction (June 13 - July 4 , 2025)
- ▶ **Phase 2** : Team Formation and Project Planning (July 11 - July 25 , 2025)
- ▶ **Phase 3** : Design and Simulation (Aug 1 - Aug 16 , 2025)
- ▶ **Phase 4** : Layout and Verification (Aug 22 - September 26 , 2025)
- ▶ **Phase 5** : Tapeout and Testing

2025 Chipathon: Blocks & Bots: An Open Chip Playground augmented with LLMs



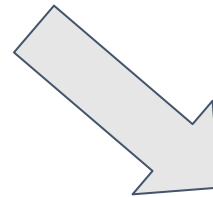
3 Tracks



Creating **MOSBius-style** chips



AI/ML enhanced Analog Automation using gLayout.



Building **Blocks for Digital Design** (new std cells lib., new memory blocks)

2025 Chipathon: Blocks & Bots: An Open Chip Playground augmented with LLMs



- ▶ Prizes:
 - Certificate for Tape out
 - Certificate for testing your Chips
 - Paths towards Code a Chip Travel Grant
- ▶ New!!! We will be introducing Challenges for each track

2025 Chipathon: Schedule

Phase 2: Team Formation and Project Planning

Week	Date	Event	Details	 Recording	 Slides	 Report
	July 05, 2025	Team Formation Deadline				
Week 28	July 11, 2025	Project Proposal Review 	-	TBA	TBA	TBA
Week 29	July 18, 2025	Advanced Topics 	• Systematic Design of Analog CMOS Circuits (Boris) • LLM RAG Finetuning & Training (Greg)	TBA	TBA	TBA
Week 30	July 25, 2025	LLM Integration 	• Tutorial: Schematic Database & Simulations (Peter) • Introduction to CACE (Tim E.) • LLM-assisted analog Layout (Saptarshi, Greg, Mehdi) • LLM assisted Digital Flow (Amro, Greg)	TBA	TBA	TBA



Phase 3: Design and Simulation

Week	Date	Event	Review Focus	 Recording	 Slides	 Report
Week 31	Aug 01, 2025	Schematic Review 	-	TBA	TBA	TBA

Goals for Week 5

- ▶ Team formation:
 - Most teams are formed
 - Last call for non-affiliated participants: Please be vocal!
- ▶ Proposals
 - 2-3 slides presentation today
 - Full Proposal this Sunday 13 July

Proposal Guideline - Key Points

- ▶ Block Diagram & Pin-out
- ▶ Schedule & Team (Feasibility)
- ▶ Testing plan (& applications depending on your track)
- ▶ Originality (can be in various aspects like design approach, design tools, chip function, architecture, ...)
- ▶ Proposal quality (as an indicator to attention to detail etc.)

Fill out the Forms!

- ▶ We will use weekly Google forms to track participation and progress. Hopefully, this will allow us to quickly address any problems.
- ▶ **Every participant** from each team is expected to complete the form each week. Links to the forms will be posted on the GitHub page.

The screenshot displays a Google Form titled "IEEE SSCS 2025 Chipathon - Week 1". The form is divided into two sections. Section 1 of 2 contains a title bar, a text prompt "Please complete the weekly forms to track your progress and report", and four input fields: "Full Name" (short answer text), "Email address" (short answer text), "Team name (if applicable)" (short answer text), and "Track" (radio button selection). Section 2 of 2 contains a title bar, a text prompt "Weekly milestones for June 13 -19", a "Description (optional)" field, a radio button selection for "June 8th online meeting attendance", a checkbox for "Chipathon 2025 chat room", a radio button selection for "Simulation environment", and a "Please list any issues you are having with the current assignments." field.

Section 1 of 2

IEEE SSCS 2025 Chipathon - Week 1

Please complete the weekly forms to track your progress and report

Full Name *

Short answer text

Email address *

Short answer text

Team name (if applicable)

Short answer text

Track *

- ☐ Track A: Creating MOSbius-style chips in GF180MCU
- ☐ Track B: Digital Building Blocks with an LLM-driven Workflow
- ☐ Track C: AI/LLM enhanced Analog Automation using gLayout
- ☐ Not sure

Section 2 of 2

Weekly milestones for June 13 -19

Description (optional)

June 8th online meeting attendance

- ☐ Attended online
- ☐ Watched the recording
- ☐ Did not attend or watch the recording

Chipathon 2025 chat room

☐ Joined element.fossi-chat.org and the chat room #chipathon-2025:fossi-chat.org

Simulation environment

- ☐ Docker environment is working
- ☐ Natively installed programs are working
- ☐ Simulation environment not working

Please list any issues you are having with the current assignments.

Long answer text

Tracking:

- ▶ The weekly form results are on a [publicly viewable tracking sheet](#).
- ▶ Use this information to collaborate and/or support other teams

IEEE SSCS 2025 Chipathon Teams																
File Edit View Insert Format Data Tools Extensions Help Accessibility																
Menus 100% 123 Defaul... 10 B I A																
A1	Track															
	A	B	C	D	F	G	H	I	J	K	L	M	N	O	P	
1	Track	Team	Time Zone UTC+/-	Members	Week 24	Week 25	Week 26	Week 27	Week 28	Week 29	Week 30	Week 31	Week 32	Week 33	Readiness Check	
2	Analog/AI	AutoMOS	UTC -5:00	Julio David Vilca Pizarro		O	O									
3	Analog/AI	AutoMOS	UTC -5:00	LuighiV	O	O	O									
4	Analog/AI	AutoMOS	UTC -3:00	Rodrigo Marin		O	O									
5	Analog/AI	Brown1		Gannon	O											
6	Analog/AI	Brown1		Martins Atilla	O											
7	Analog/AI	Brown1	UTC +8:00	Ray		O	O									
8	Analog/AI	Capibara	UTC -6:00	Jorge Alejandro Juarez Lora			O									

<Team Name>

Team members:

Leader name <fossi-chat name>

Team member name 1

Team member name 2

Team member name 3

Team member name 3

Team member name 4

Team member name 5

Team background

- Academic Experience
- Work Experience

<Team Name>

Project information

- Goal (2 sentences)
- Design - High Level Proposal (2-3 sentences + Image and/or link to details)
- Application (1 sentence)
- References

Questions, Suggestions, Doubts?

-

Analog/AI Track Project Reviews (6)

- AutoMOS
- Capybara-Analog
- GenYZ team
- Mahowalders
- Silicon Levelling
- Time Transcenders
- Watt the Heck

AutoMOS

Team members:

Luighi Viton-Zorrilla <LuighiV>

Julio David Vilca Pizarro

Rodrigo Marin

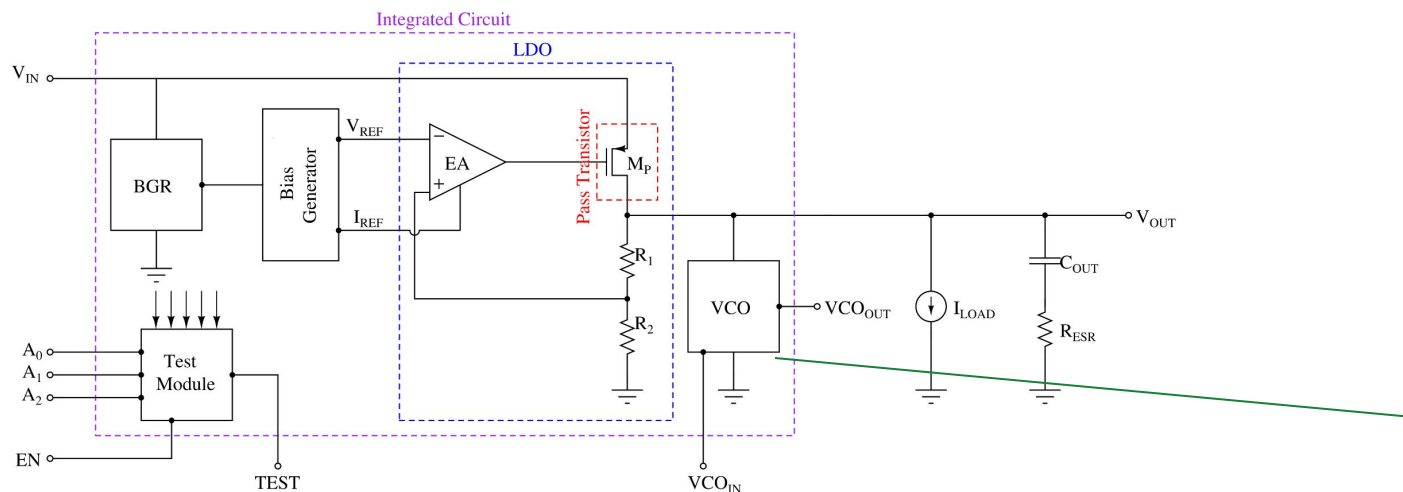
Team background

- Academic Experience:
 - 2 MEng in Biomedical engineering
 - 1 BSc in Electronic Engineering
- Work Experience
 - 2 members with +2 years of experience as analog IC designer
 - 1 member with +1 year of experience as analog IC designer

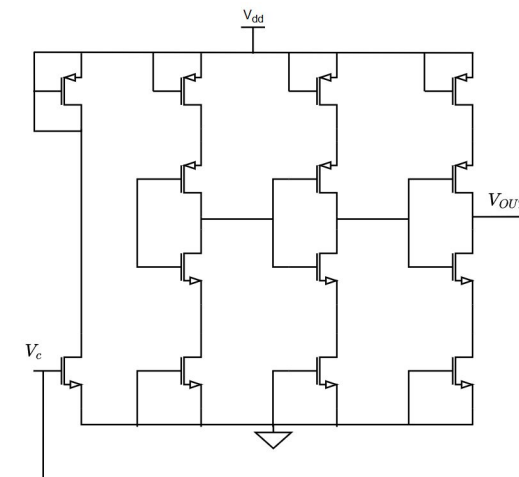
AutoMOS

Project information

- Goal:
 - We aim to build a basic power management chip (DC-DC converter) implementing a LDO as its core and including a VCO as an integrated load.
 - Contribute to the glayout repository with the building blocks designer for the implementation.
- Design - High Level Proposal:
 - 1 CMOS Bandgap reference (1.2V) and 1 bias generator for current (PTAT, ZTC) and reference voltage (1V, 0.5V)
 - 1 LDO (1.8V) with 0.2V of dropout
 - 1 VCO based on current-Starved Ring oscillator (1.8V) for 2.4 GHz



Block diagram for VCO



- Application: This circuit is designed to provide regulated voltage rails for other system components, including MCUs, RF modules, ADCs, and others.
- References:
 - https://www.seas.ucla.edu/brweb/papers/Journals/BR_SSCM_3_2021.pdf
 - <https://www.ti.com/lit/an/slva118a/slva118a.pdf>
 - <https://ieeexplore.ieee.org/document/7285037>
 - <https://www.mdpi.com/1996-1073/13/11/2986>
- GitHub repository: <https://github.com/AutoMOS-project/AutoMOS-chipathon2025/blob/main/docs/proposal.md>
- Proposal detail: <https://docs.google.com/presentation/d/1TvIR2NhGDkT4yeVhsB3TjcvvHDocXAfSwPzngiaAozw>

Questions, Suggestions, Doubts?

- Do you believe the scope of the project is ok or do we need to reduce it considering the available time we have?

Lead:

Team: Brown1

Project proposal

Team: Capybara-Analog

Team members:

Jorge Alejandro Juarez Lora <alejandrojuarezlora>

Raul Perez Nuñez <RalPerez>

Gonzalo Sebastian Torres Chavez <salieri_117>

Abrahan Alejandro Salazar Hernandez

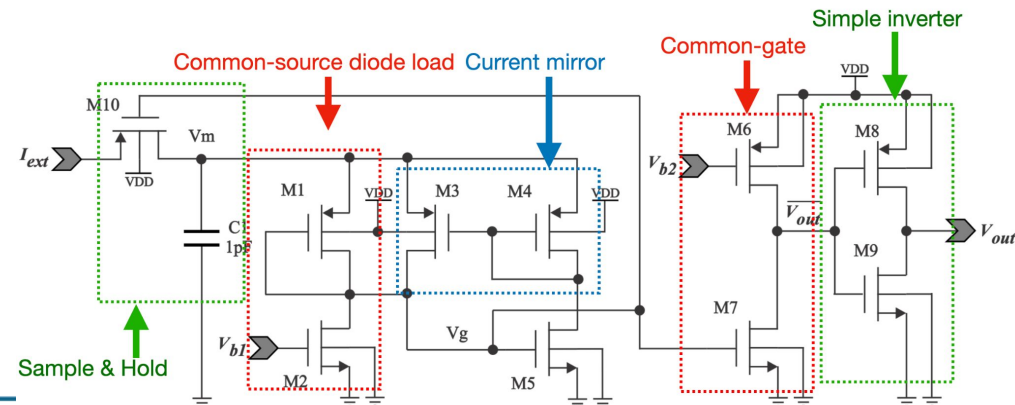
Team background

- Academic Experience:
 - 1 PhD in Computer Science
 - 2 MSc in Computer Engineering
 - 1 BSc in Bionics Engineering

Project Information

Team: Capybara-Analog

- Goal:
 - Use gLayout to craft the layout automatization of spiking neural networks building blocks
 - Collaborate into Migrate some of the components of the gLayout component plan from SKY130 to GF180MCU to assemble them as a spiking neuron
- Design - High Level Proposal:
 - Class B push-pull, Common source amplifier, current mirror, sample and hold and transmission gates at GF180MCU
 - Layout of a spiking analog neuron using these cells and gLayout
- Application:
 - Create a framework based in glayout, GF180MCU, SKY130, and open source tools to craft analog spiking neural networks from Text interface to GDSII
- References:
 - <https://doi.org/10.3390/math12142267>
 - <https://doi.org/10.3389/fnins.2021.627221>
 - <https://doi.org/10.1109/ICCAD57390.2023.10323720>
 - <https://doi.org/10.1038/s41586-024-08253-8>
- Github Repository: <https://github.com/AlejandroJuarezLora/Team-Capybara-Analog/tree/main>



GenYZ Team

Team members:

Keisya Andretta Basoeki <keisyaandretta>

Aleta Edna Jessalyn

Angelina Wahyuni

Abdillah Aziz

Infal Syafalni

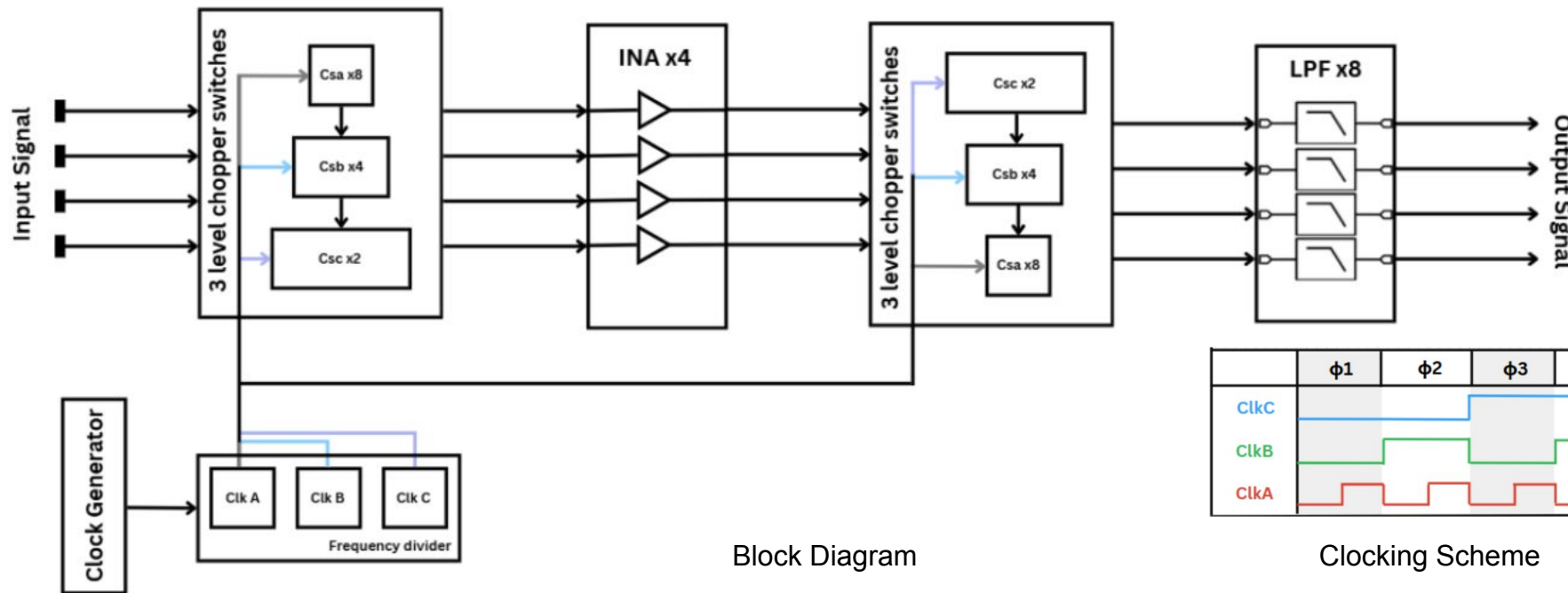
Team background

- Academic Experience
 - 3 undergraduate students (3rd year) ITB
 - 1 researcher ITB
 - 1 lecturer ITB
- Work Experience
 - Researcher > 2 years in analog IC design & testing
 - Lecturer > 5 years in digital IC design

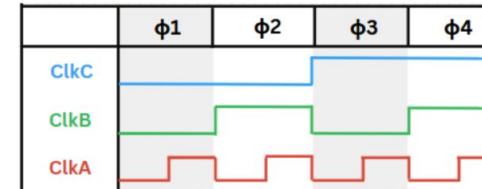
Gen YZ Team

Project information

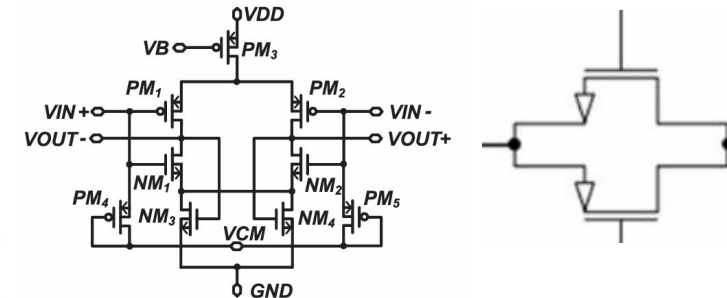
- Goal:
 - Low mismatch 4 channels instrumentation amplifier using LLM. Scalable for more channel for future.
- Design - High Level Proposal
 - Consist of 4 differential amplifiers, 14 chopper switches in 3 stages, 1 digital frequency divider for clocking scheme, and 4 switch cap LPFs



Block Diagram

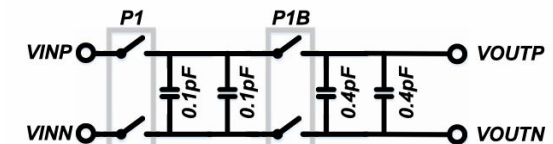


Clocking Scheme



Full differential amplifier

Chopper switch



Low Pass Filter

Gen YZ Team

Project information

- Application
 - Frontal lobe EEG measurement
- References
 - Tao Tang, Jeong Hoan Park, Lian Zhang, Kian Ann Ng, and Jerald Yoo. "Group Chopping: An 8-Channel, 0.04% Gain Mismatch, 2.1 μ W 0.017 mm² Instrumentation Amplifier for Bio-potential Recording". IEEE Journal of Solid-State Circuits, vol. 16, no. 3, pp. 1061–1071, Jun. 2022, doi: 10.1109/TBCAS.2022.3166513
 - WhaleTeq Medical Device Manufacturer. "CMRR Test Principle and Method." Accessed from <https://www.whaleteq.com/en/app/view18-cmrr-test-principle-and-method> at July 9th, 22.19 PM UTC+7.00.
 - S. Yazicioglu, T. Torfs, P. Merken, H. Kim, J. Penders, R. F. Yazicioglu, and C. Van Hoof, "A 0.5 V 2.1 μ W EEG acquisition IC with differential and common-mode active DC offset rejection," IEEE Journal of Solid-State Circuits, vol. 57, no. 4, pp. 1061–1071, Apr. 2022, doi: 10.1109/JSSC.2022.3161704.

Questions, Suggestions, Doubts?

- Need suggestion which cap to use in LPF?

Team: Mahowalders (Team M)

- Members:

- Sudhansu Mishra (Lead: @sud-ana)
- Quentin Halbach
- Saptarshi Ghosh

Accepting new members

- Team Experience

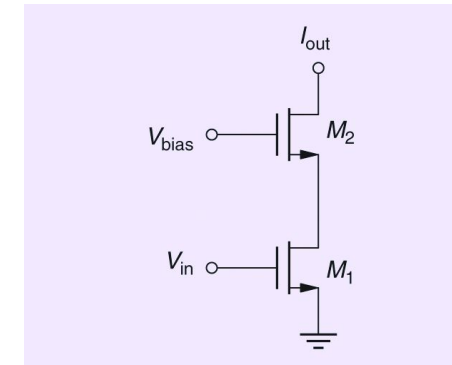
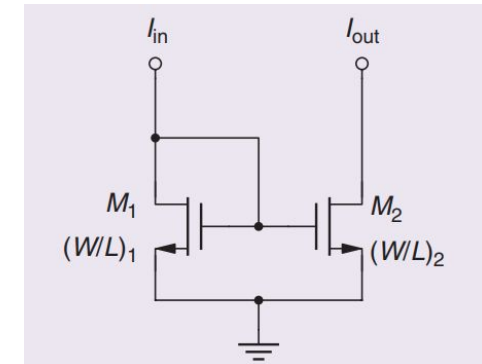
- 1 PostDoc + 1 PhD + 1 Btech Engineer
- Minimal experience in Analog IC design
- 2 Participants of CP'24

- Design Proposal Title: Towards Precision Current Mirrors/Bias Generators

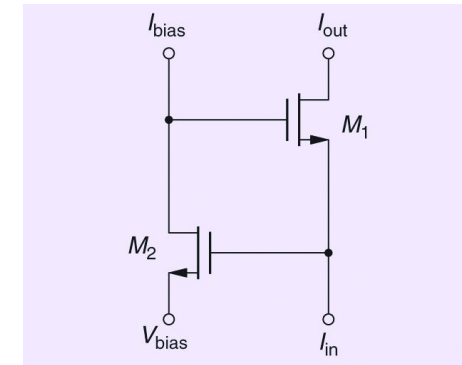
- Design Goal:

- To design and implement a regulated current mirror source capable of supplying stable bias currents for analog and mixed-signal cells.
- Using gLayout based *python2GDS* generation of a high output impedance bias generator (Biasgen) across varying loads and process conditions

Copied from Pretl, Harald, and Matthias Eberlein. "Fifty nifty variations of two-transistor circuits: A tribute to the versatility of MOSFETs." *IEEE Solid-State Circuits Magazine* 13.3 (2021): 38-46 and [references therein](#)

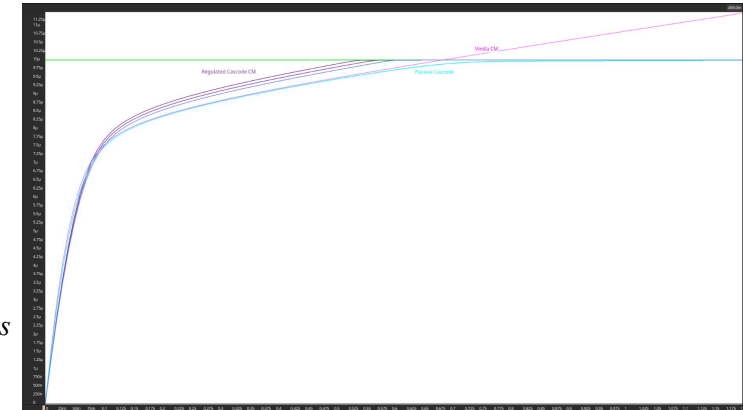


/



Team: Mahowalders (Team M)

- Design:
 - Using 3-different parametric current mirror blocks, two OTAs, and a custom cmirror branch, we plan to design an integrated cell
 - Vanilla CM (vcm) ● Self Biased Common-Source CM (sb cm) ● Regulated Cascoded CM (rc cm)
 - Outputs available on chip-pins will be tested with external interface (Shunt resistors) and sense-ADCs to measure current precision.
 - Cross-correlation between three branches under different load and process conditions will be done in testing phase
- Application
 - Bias Current /Reference Generation
 - Op-Amp or DAC/ADC design sub-circuit
 - Sensor Front-end Interfaces
- References
 - Michal, Vratislav. "Regulated Cascode Current Mirror with Improved Voltage Swing." 2022 International Conference on Applied Electronics (AE). IEEE, 2022. DOI : 10.1109/AE54730.2022.9920096
 - Garde, M. Pilar, et al. "Wide-Swing Class AB Regulated Cascode Current Mirror." 2020 IEEE International Symposium on Circuits and Systems (ISCAS). IEEE, 2020.
 - Pretl, Harald, and Matthias Eberlein. "Fifty nifty variations of two-transistor circuits: A tribute to the versatility of MOSFETs." *IEEE Solid-State Circuits Magazine* 13.3 (2021): 38-46 and [references therein](#)
 - GF180 PadFrame Document: [PADFrame - Chipathon 2023](#)



[CP'24 Presentation](#)



[GitHub repository](#)



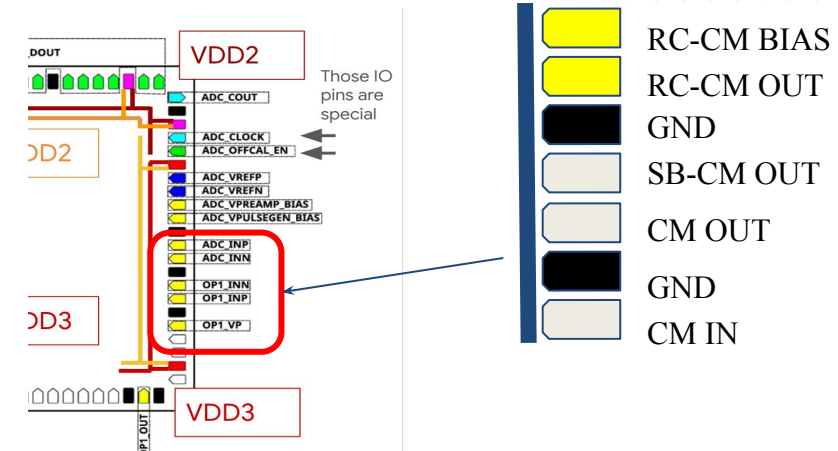
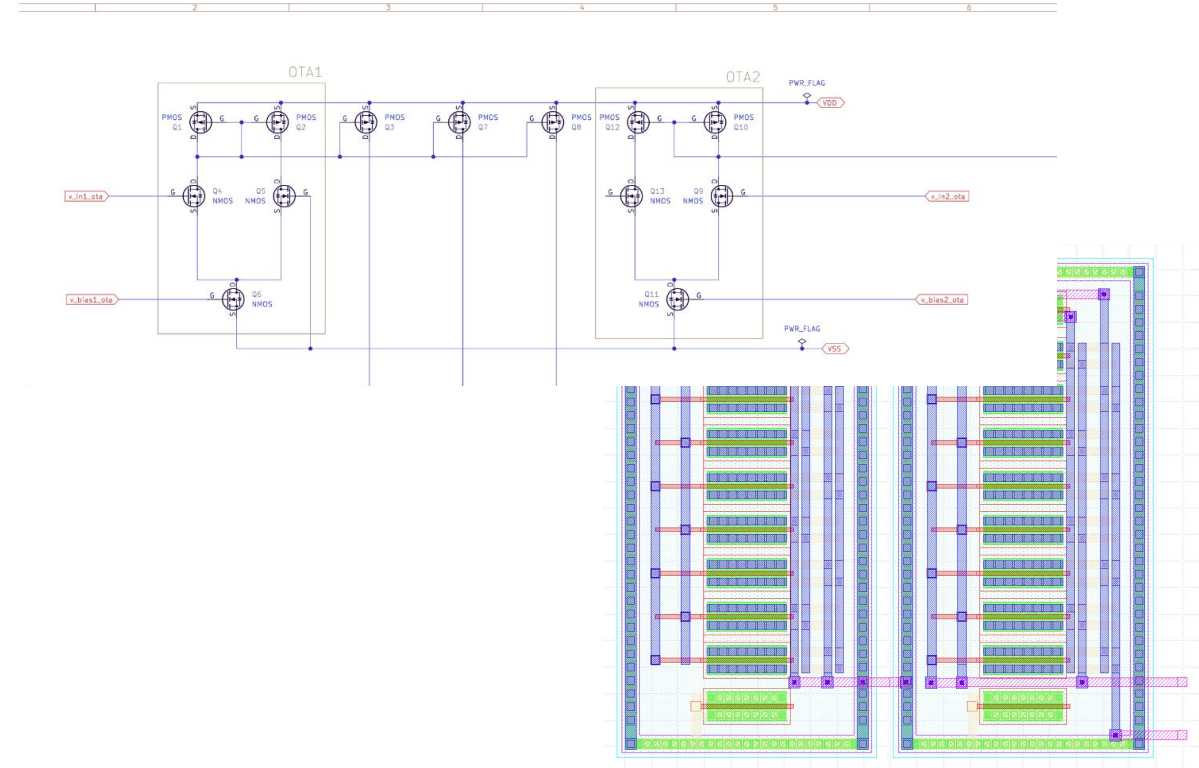
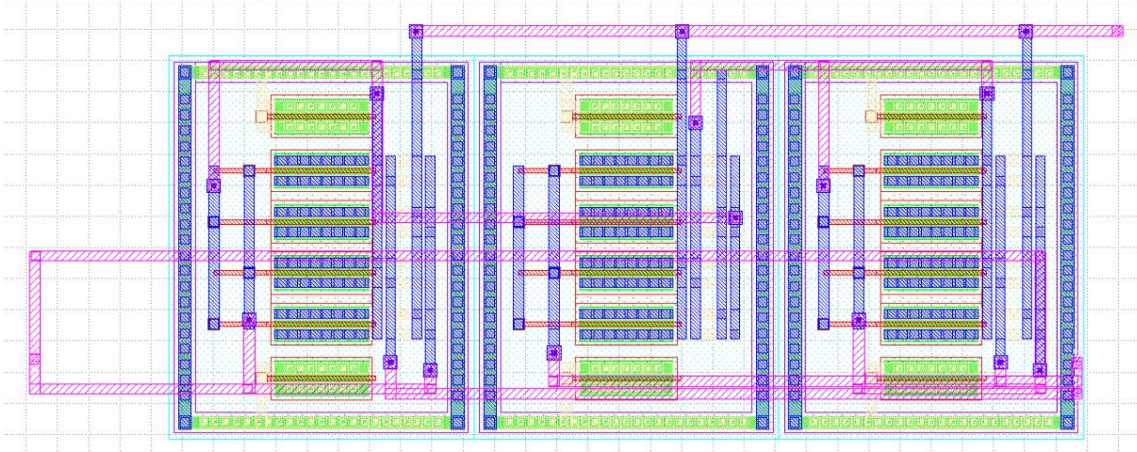
[Proposal @
GitHub repository \(WIP\)](#)



[Proposal @
Google Docs \(WIP\)](#)

Team: Mahowalders (Team-M)

- Questions
 - Applications -Integration in larger design
 - Temperature compensation
- Doubts
 - Current Source: external or OTA or simple PolyRes with fixed reference
 - Mux'ed Output ?
 - SKY130 to GF180 transition



Team: On Chip Team

Lead:

Project proposal

Lead:

Team: SAMA lab

Project proposal

Team: Silicon Souls

Lead: Subham Pal
Member- Ekta Singh

Project proposal https://docs.google.com/presentation/d/1ZdI_hSxEF5d12hDnMMYXRdIQuC9EC54WVXz7nvXz4g8/edit?usp=sharing

Silicon Levelling

Team members: Shubham Kumar
(shubham-analog)

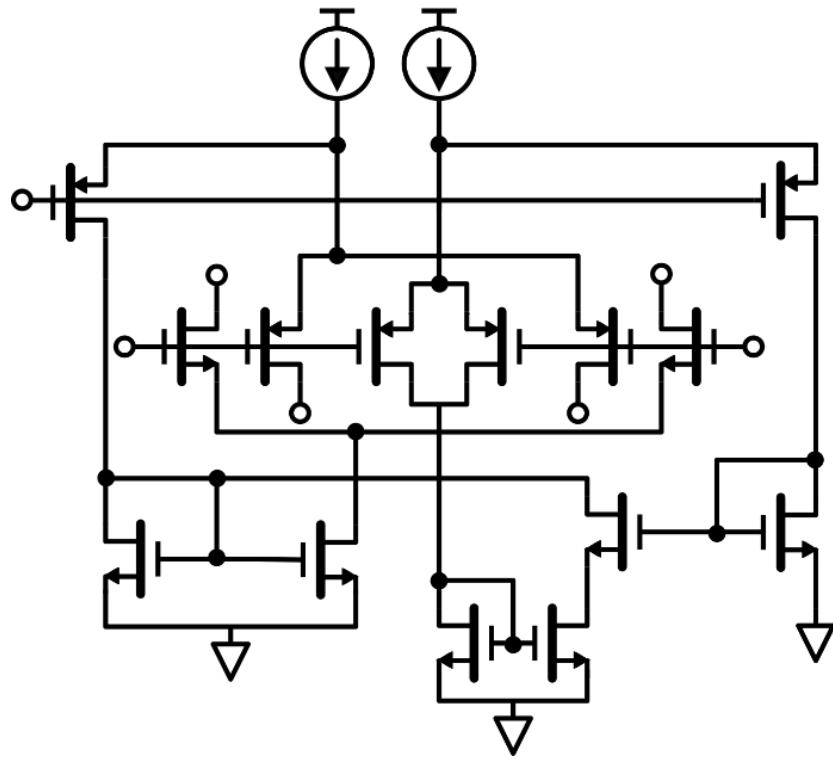
Team background

- Academic Experience
 - Dual Degree B.Tech-M.Tech from Indian Institute of Technology(IIT), Kharagpur, 2024
 - Major: Instrumentation and Integrated Electronics, Dept. of Electrical Engineering
 - Minor: Computer Science and Engineering
- Work Experience
 - Analog Design Engineer at Texas Instruments, Bangalore, India (2024 - Present)

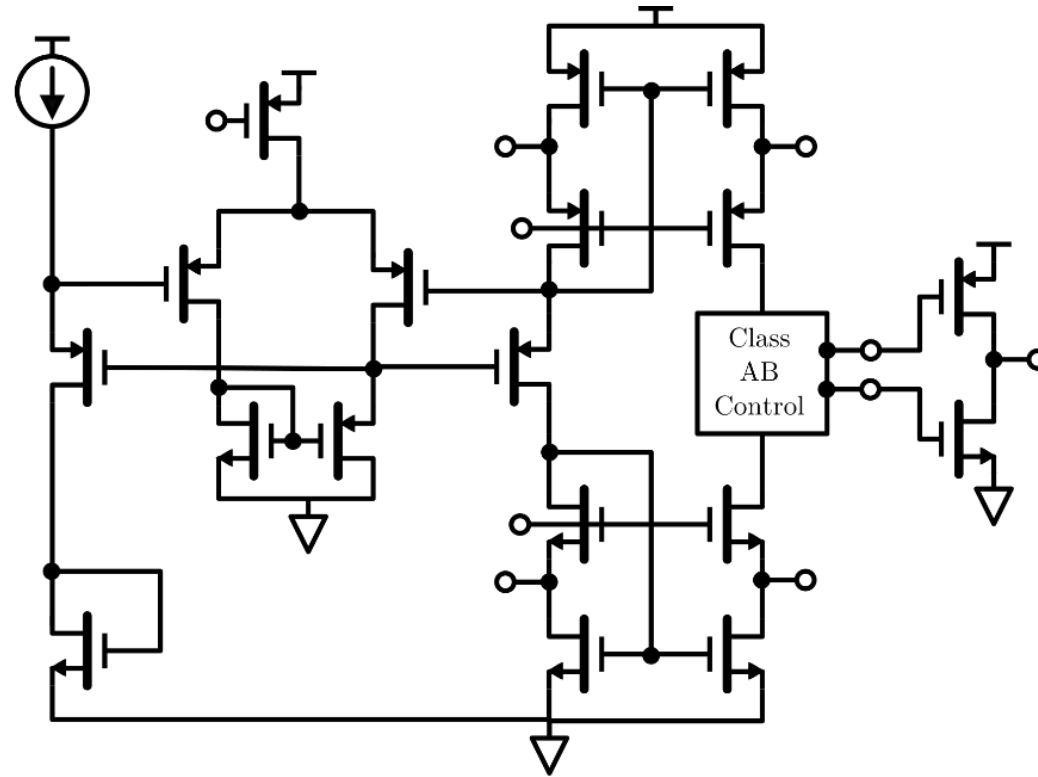
Silicon Levelling

Project information

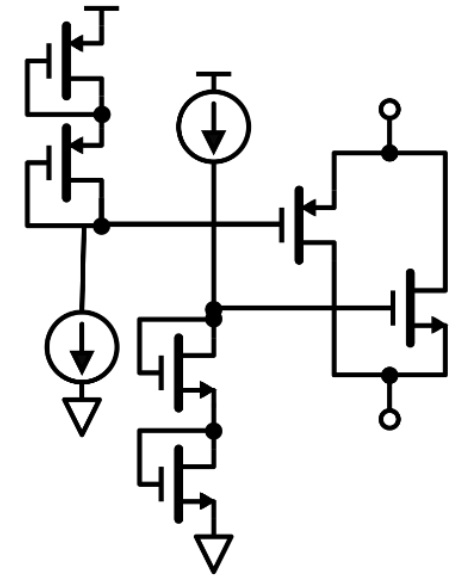
- Goal: To build a Rail-to-Rail Class-AB OTA. To be able to contribute new pcells to Glayout, such as,
 - Rail-to-Rail input tail current switching block
 - Minimum selector block
 - Rail-to-Rail Input Stage
 - Floating current source based Intermediate block
 - Class-AB Biasing block
- Design - A complete Rail-to-Rail Class-AB OTA on 3.3V power supply. Circuit diagram on next slide. More details on specs to be added.
- Application - To be used as a generic all purpose OTA, eg. Buffers, other fun stuffs(put some caps and make a oscillator :P), etc.
- References - Ivanov, Vadim V., and Igor M. Filanovsky. Operational amplifier speed and accuracy improvement: analog circuit design with structural methodology. Vol. 763. Springer Science & Business Media, 2005.
-



[Left]: Rail-to-Rail Input Stage with
Current Switching Scheme
(augmented with minimum selector
circuit)



[Center]: Symmetric Intermediate
Stage for folding, with Floating Current
Source and Class-AB Bias



[Right]: Class-AB Biasing
Scheme

Time Transcenders

Team members:

Vasil Yordanov
<vasil_yordanov>
Hari Preeth
Shaikh Shoeb Dawood
Yuan Yancey Labay

Team background

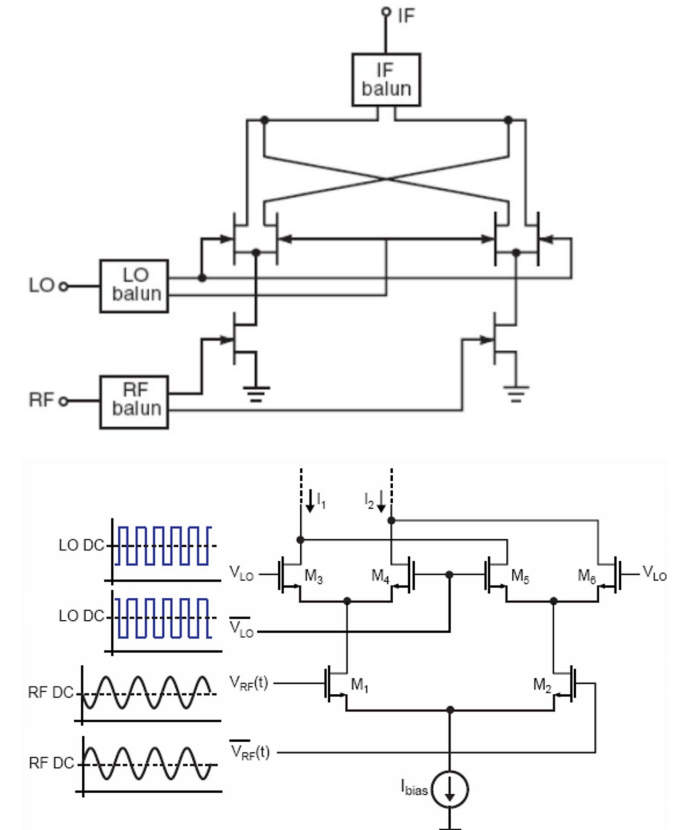
- Academic Experience
- Work Experience

Time Transcenders

Project information

- Implement a Gilbert cell double-balanced mixer in GF180MCU-D.
If time permits also implement Balun devices
- Aim for a mixer for WiFi front-end applications, working at 2.4GHz.
Layout is to be done using the gLayout framework
(programmatically), which is novel.
- References:
 - https://github.com/Landflier/Chipathon_2025_gLayout/blob/main/doc/TimeTranscenders_Proposal.pdf

Questions, Suggestions, Doubts?



Watt the Heck

Team Members: David Li and Kireeti Kompella

Goal: Automating layout generation for large-scale spiking neural-network-on-chip using the silicon control rectifier (SCR) architecture.

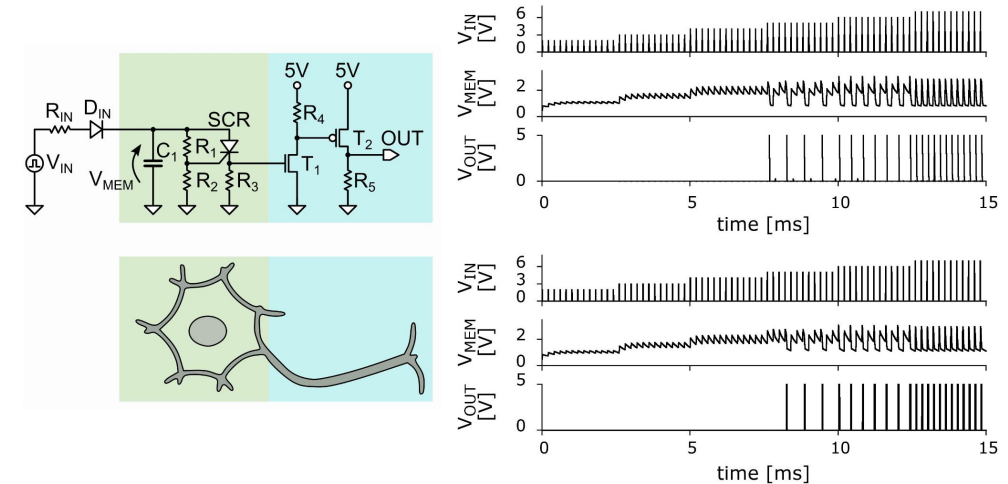
Design:

- Biologically accurate (refractory period and voltage spiking rate modulation) spiking neural neuron implementation with SCR from [1] adapted to CMOS components (reference design shown in Fig.)
- AI/LLM assisted analog design to support reconfigurable large scale integration and optimization of chained spiking neural network layout.

Application: Creating neuromorphic computing systems and building large scale optimized spiking neural networks.

Reference:

- [1] [An ultra-compact leaky-integrate-and-fire model for building spiking neural networks | Scientific Reports](#)
- [2] [Biologically Relevant Dynamical Behaviors Realized in an Ultra-Compact Neuron Model](#)



Digital Building Block Track Project Reviews (19)

- @meGanesha
- BitSamaMirai
- CCSL
- CellTroopers
- CHIPABROS
- ChipiChapa
- Chipster
- Chipstronics

- Inyeon
- Mission Silicon
- Nawiatsob
- Setup & Hold
- Siliclusters
- SILICON_BBI1
- SILICON_BBI2
- SiliconB
- Symbelleuit
- VoidWalkers

@meGanesha

Team members:

Irfan Nurhakim Hilmi <@irfannhlm:matrix.org>

Randy Revaldo Pratama

Nur Dawam Abdan Syakuro

Avila Khadhibyan

Team background

- Academic Experience

All just finished 3rd year in Electrical Engineering, Institut Teknologi Bandung (ITB), Indonesia

- Work Experience

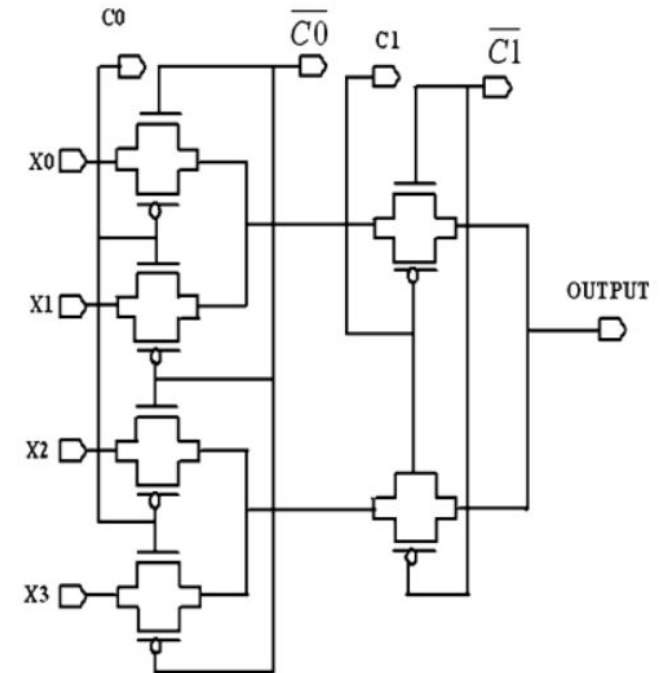
None

Project information

- **Goal**
Create efficient pass transistor logic (PTL) based multiplexers as standard cells. Provide full characterization for open-source library integration.
- **Design - High Level Proposal**
The plan is to design 4-1 MUX using transmission gates. The target is less area with similar (or better) characteristics than standard multiplexers. Example 4-1 MUX design can be seen on the side.
- **Application**
Original idea is to use these cells for area-efficient barrel shifters, but can definitely be used for other macro cells.
- **References**
M. Mishra and S. Akashe, "High performance, low power 200 Gb/s 4:1 MUX with TGL in 45 nm technology," *Appl. Nanosci.*, vol. 4, no. 8, pp. 889–895, Dec. 2014, doi: 10.1007/s13204-013-0206-0

Questions, Suggestions, Doubts?

- Any suggestions on effective task allocation?
- Is it better if each person work on different variations, or stay with 4 people for one cell?
- Possibility of adding demux design if too simple?



Lead:

Team: BK-G9

Project proposal

BitSamaMirai

Team members:

Leader name <fossi-chat name>

- Reinhard Iven
- Keerthana N
- Roi Victor Roberto
- Kana Takizawa

Team background

- Academic Experience
 - 2nd - 4th year of undergraduates
 - Digital design using VHDL/Verilog and FPGA
 - iverilog
- Work Experience
 - Digital design verification

BitSamaMirai

Project information

- **Goal**

To design 3-input XOR for gf180mcu_osu_sc_gp9t3v3 and gf180mcu_osu_sc_gp12t3v3

- **Design - High Level Proposal (2-3 sentences + Image and/or link to details)**

- **Application**

XOR logic is widely used to generate parity bits.

- **References**

Questions, Suggestions, Doubts?

-

CCSL

Team members: Jinane Bazzi <@jinanebazzi>
Bayan Jaber

Team background

- Academic Experience:

- Bayan Jaber is a Master's student in Electrical and Computer Engineering at KAUST.
- Jinane Bazzi is a Ph.D. student in Electrical and Computer Engineering at KAUST.

CCSL

Project information

- Goal:
 - Develop a memory compiler specifically designed for GF180MCU technology, capable of generating SRAM blocks of arbitrary sizes efficiently. The compiler will streamline SRAM generation, characterization, and validation processes.
- Design - High Level Proposal:
 - The proposed compiler will identify available SRAM macros within the GF180MCU library and optimize macro combinations based on specified power and area constraints.
- Application:
 - significantly reduces design complexity, accelerates memory integration into larger digital systems, and facilitates rapid prototyping and chip validation post-tapeout.
- References:
 - https://gf180mcu-pdk.readthedocs.io/en/latest/IPs/SRAM/gf180mcu_fd_ip_sram/cells/gf180mcu_fd_ip_sram_sram64x8m8wm1/gf180mcu_fd_ip_sram_sram64x8m8wm1.html
 - <https://colab.research.google.com/drive/1U5-XR46EepGHZvS8sSF6omnSpjU8DPMg?usp=sharing>

CellTroopers

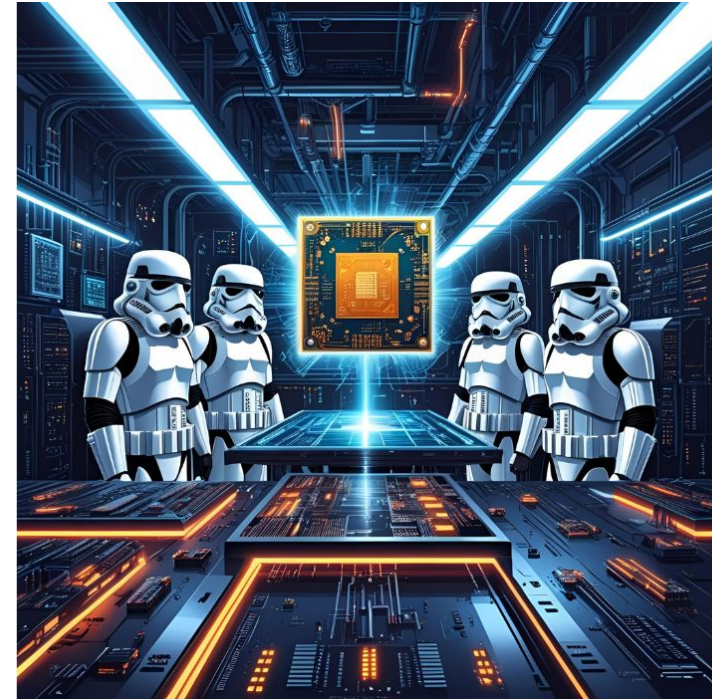
Team members:

Lucas Franck <ldfranck>

Matheus Souza <matheussouza>

William Orlato <williamorlato>

UTC -03:00 (Brazil)



Team background

Lucas: Electrical Engineer from the University of São Paulo (USP), Chipalooza finalist.

Matheus: Electrical Engineer from the Federal University of São Carlos (UFScar), ChipInventor leader.

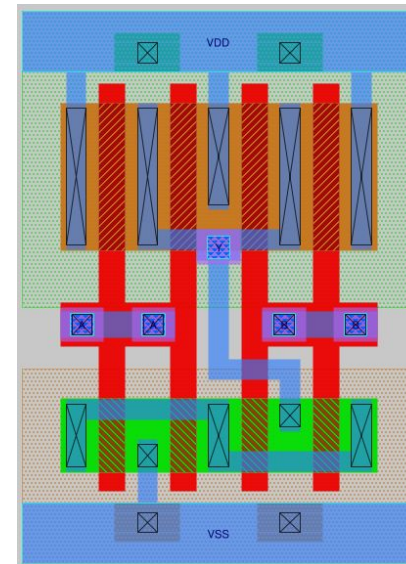
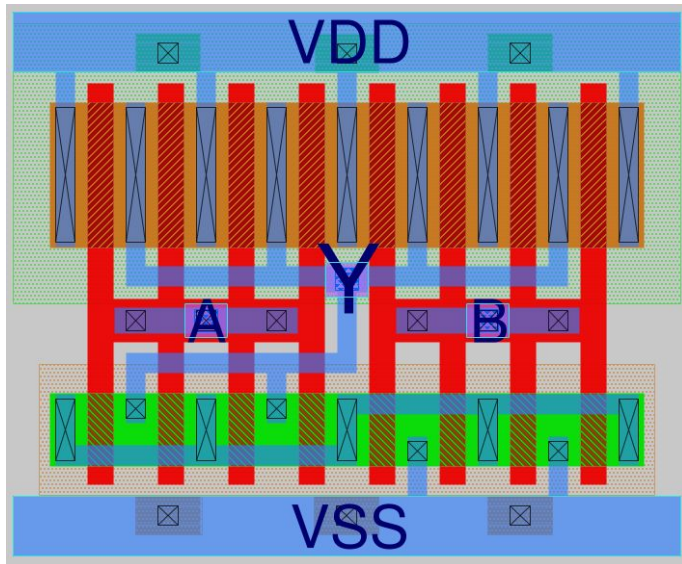
William: Electrical Engineer from the University of São Paulo (USP), Chipalooza finalist.

CellTroopers

Project Information:

Goals: contribute to the ongoing effort to expand OSU's std cell library
test new open-source tools for cell characterization

Proposal: start with 2-input NAND gates with drive strength x2 and x4 as a warm-up
continue to work on Tim's std cell list as we complete the designs



CHIPABROS

Team members:

Karunia Muda Kurniawan <Karunia Muda Kurniawan>

M. Ahza Bayanaka Wijanarko

Akmal Hafizh Musyaffa

M. Azfar A. Yusup

Team background

- Academic Experience

All of the members just finished 3rd Year as an Electrical Engineering Undergraduate with a basic to intermediate knowledge of ASIC and IC Design Open-Source tools

- Work Experience

We implemented the layout for a complete RISC-V processor with integrated SRAM, leveraging the open-source sky130 PDK and a full toolchain of OpenLane and OpenRAM.

CHIPABROS

Project information

- **Goal**

We aim to extend the OSU library with **AOI33**, **OAI33**, and **MUX4** cells to improve area, delay, and routability in complex designs.

- **Design - High Level Proposal**

MUX4 selects one of four inputs (x_0, x_1, x_2, x_3) based on control signals C_0 and C_1 , outputting M .

AOI33 computes $y = \sim((x_0 \cdot x_1 \cdot x_2) + (x_3 \cdot x_4 \cdot x_5))$, combining two 3-input ANDs with a NOR.

OAI33 computes $y = \sim((x_0 + x_1 + x_2) \cdot (x_3 + x_4 + x_5))$, combining two 3-input ORs with a NAND.

Physical Design Note: For all gates, the Pull-up (PUN, using pMOS) and Pull-down (PDN, using nMOS) networks will be implemented as complementary CMOS duals to ensure correct logic operation and minimize static power.

- **Application**

These cells are widely applied in control logic, arithmetic units, and multiplexing circuits to enable efficient and scalable digital designs.

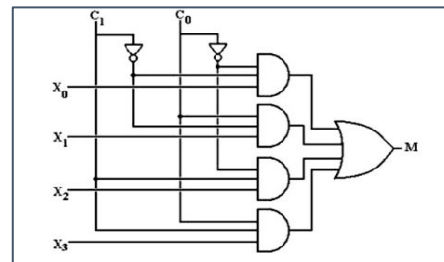


Figure 1 MUX4

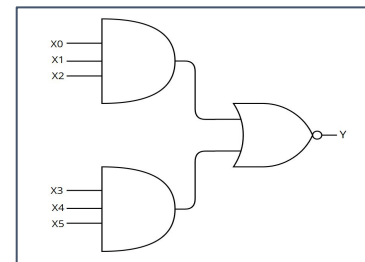


Figure 2 AOI33

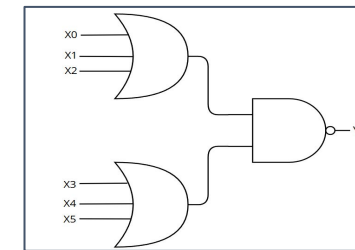


Figure 3 OAI33

Lead:

Team: Chipathon - 02

Project proposal

ChipiChapa

Team members:

Rafi Ihsan Alfathin <@rafi_ihsan_alfathin:matrix.org>

Adrian Sami Pratama

Muhammad Nabil Raihan

Saputra Yudika Marpaung

Team background

- Academic Experience

All team members just finished 2nd year in Electrical Engineering, Bandung Institute of Technology, Indonesia.

- Work Experience

No experience at all.

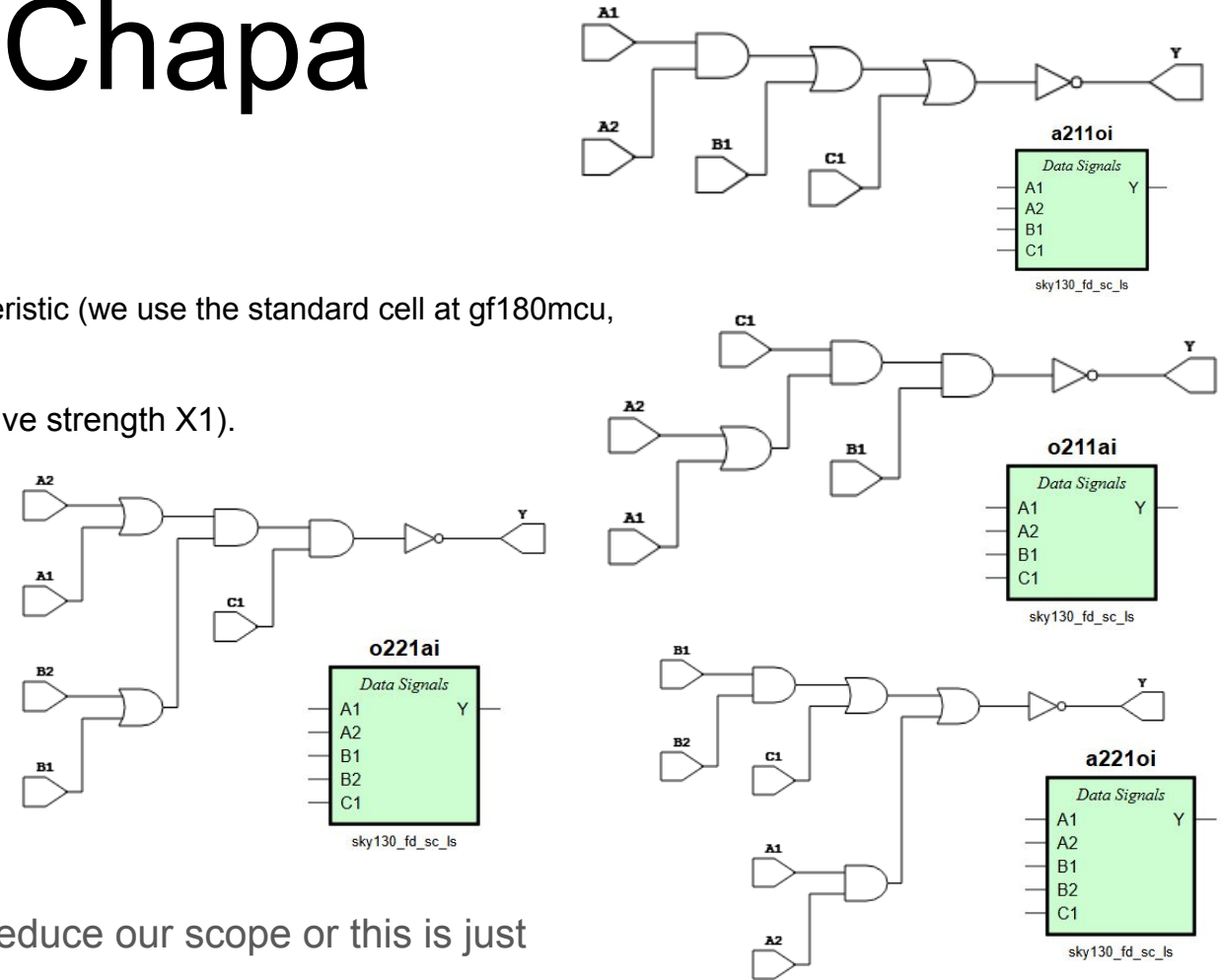
ChipiChapa

Project information

- **Goal**
Build AOI and OAI standard cells (9 track version) with an efficient characteristic (we use the standard cell at gf180mcu, skywater130, and research paper for further references)
- **Design - High Level Proposal**
The plan is to design AOI211, OAI211, AOI221, and OAI221 (9 tracks, drive strength X1).
- **Application**
Can be used for arithmetic units, multipliers, multiplexer, etc.
- **References**
<https://gf180mcu-pdk.readthedocs.io>
https://github.com/stineje/globalfoundries-pdk-libs-gf180mcu_osu_sc
<https://github.com/google/skywater-pdk>
- **Our Team Github Repo (with proposal):**
https://github.com/AdrianSPratama/SSCS-Chipaton-2025_ChipiChapa

Questions, Suggestions, Doubts?

- Given our academic background, did we need to expand/reduce our scope or this is just enough?
- What is a good work distribution, is 1 gate for 1 person good?
- Need more references/tutorial for whole design process and to make an efficient gate beside transistor sizing



CHIPSTER

Team members: Ade Irman Budi Hendriawan
Najmi Azzahra
William
Rifki Afriadi

Team background

- Academic Experience

Irman: Undergraduate (Bandung Institute of Technology, Indonesia)

Najmi: Undergraduate (Bandung Institute of Technology)

William: 2nd Year Undergraduate (Bandung Institute of Technology)

Rifki Afriadi: 2nd Year Undergraduate (Bandung Institute of Technology)

- Work Experience

Hardware Accelerator for AI, Number Theoretic Transform, Mixed Signal, Layout Std Cells and Chip.

CHIPSTER

Project information

- **Large Language Model (LLM) for RTL and Standard Cells Generation for GF180 PDK, From Prompt to Chip**
- Tapeout if Possible = Quantum Fourier Transform (QFT) 2 Qubits - if possible generated from LLM we will make, we will be started from smaller design first
- Dataset = 690k+ RTL Prompt and Verilog Generation for fine tuning the LLM
- <https://arxiv.org/abs/2406.06549> (LLM for Std Cells Gen, Ren Haoxing, NVIDIA, 2024) - **HOT TOPIC**
- Magic Layout formats will be used as datasets for LLM for Std Cells Generation

Questions, Suggestions, Doubts?

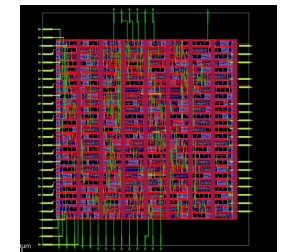
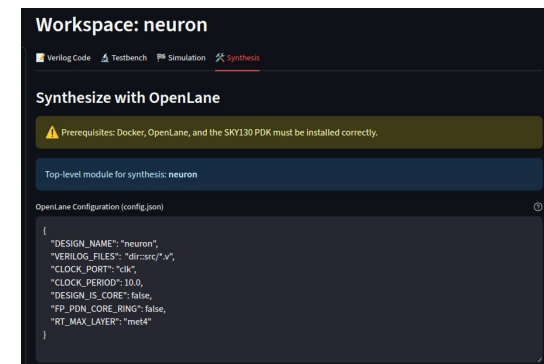
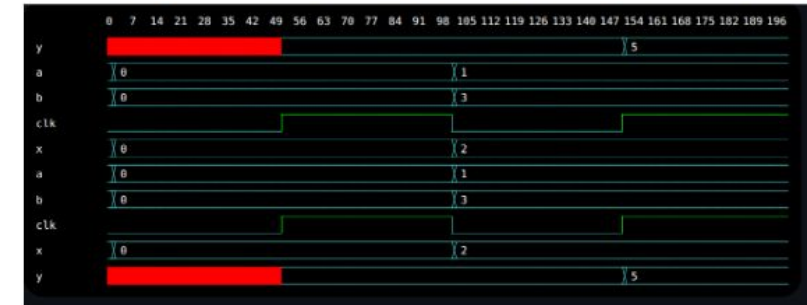
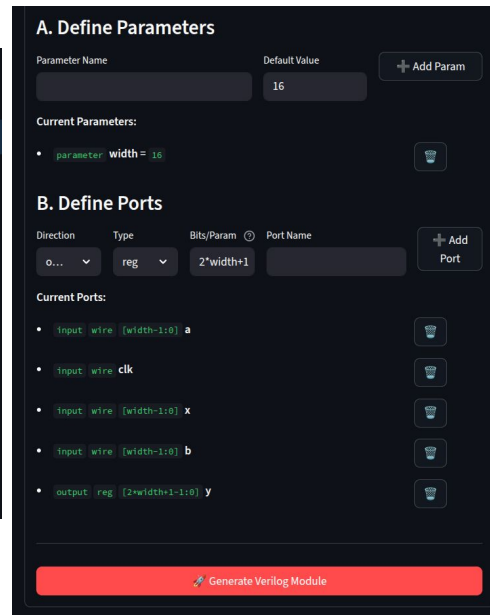
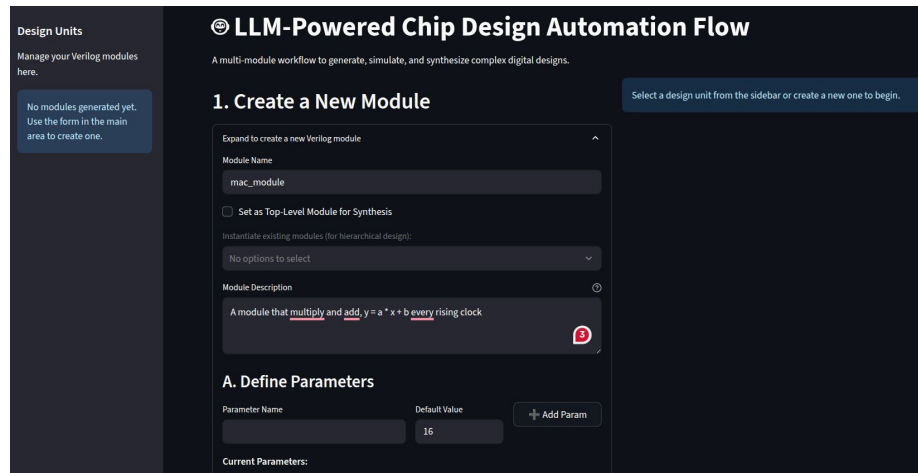
- Doubt: The .mag files we had were so small as a dataset (<10 combinations), another possibilities like using .gds files also considered.
- Doubt: Fine tuning LLM to learn with .mag datasets
- Doubt: QFT will be finished on time

CHIPSTER

Current Progress

Large Language Model (LLM) for RTL Generation (Module and Testbench)

- **PROMPT to CHIP in One Workflow**, Supports hierarchical design, CLOSED LOOP (error will be an input to LLM to generate new one)
- Neural Network (simple neuron 2 -> 1), called the MAC Module (multiply and add)
- **Next : Fine tune model and create for QFT**



MAC Module

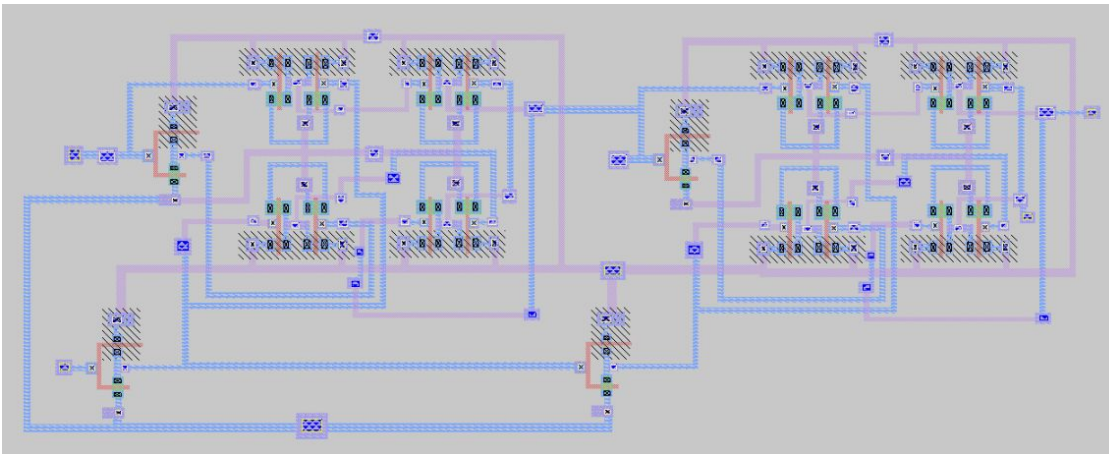
51 (Do similar for Neuron, called the MAC)

CHIPSTER

Current Progress

Continuation (+) : Large Language Model (LLM) for Standard Cells Creation, From Prompt to .mag

- Layout, DRC, and LVS of Inverter, NAND, AND, NOR, OR, XOR, MUX, D-Latch, DFF have created using Sky130 PDK. Operating Frequency of DFF is 1.075 GHz with size 50.4 μm x 19.15 μm .
- Next : Train the model to adapt with .mag and created new one with smaller size, shortest route, and higher freq.



```
Open [icon] inverter_common.mag Save [icon] [icon] [icon] [icon]
1 magic
2 tech sky130A
3 timestamp 1743223184
4 << nwell >>
5 rect -60 -65 115 175
6 << nmos >>
7 rect -15 -200 35 -185
8 << pmos >>
9 rect -15 0 30 15
10 << ndiff >>
11 rect -15 -150 35 -140
12 rect -15 -170 -5 -150
13 rect 25 -170 35 -150
14 rect -15 -185 35 -170
15 rect -15 -215 35 -200
16 rect -15 -235 -5 -215
17 rect 25 -235 35 -215
18 rect -15 -240 35 -235
19 << pdiff >>
20 rect -15 45 30 55
21 rect -15 25 -5 45
22 rect 20 25 30 45
23 rect -15 15 30 25
24 rect -15 -15 30 0
25 rect -15 -40 -5 -15
26 rect 20 -40 30 -15
27 rect -15 -45 30 -40
28 << ndiffc >>
29 rect -5 -170 25 -150
30 rect -5 -235 25 -215
```

CHIPSTRONICS

Team members:

Johan Fernandes <@Johan_Fernandes@matrix.org>

Muhammad Farhan

Mohith V

Mohd Zeeshan

Team background

- Academic Experience:

johan: 3rd year Undergraduate (India Goa College of Engineering)

Farhan: Undergraduate (Indonesia Institut Teknologi Sepuluh Nopembe)

Mohith V: PhD (India Vellore Institute of Technology)

zeeshan: Undergraduate (India MJCET)

- Work Experience:

No Professional Experience

CHIPSTRONICS

Project information

- Developing a thermometer-to-binary encoder using the full adders.
- This proposal outlines the development of a **scalable and power-efficient Thermometer-to-Binary (T2B) Encoder IP core** for mixed-signal ASIC designs. Using highly optimized full-adder standard cells, the design aims to provide a robust solution for converting ADC outputs into binary.
- The primary benefit of using full adders for this thermometer-to-binary encoder in VLSI is the **efficient reuse of standard cell arithmetic blocks to create a robust mixed-signal interface component within a PDK.**
- <https://ieeexplore.ieee.org/document/11012450>

Questions, Suggestions, Doubts?

- How to actually do the layout part in the magic for developing a standard cell as many standard cell are tightly packed do we have import the pcell from the schematic or make the custom fet's in layout?
- How to characterize the circuit as standard cell using open-source tools? (if any)

Lead:

Team: Chipzter

Project proposal

DIGIMONS

Team Members:

- Wilson Vo (team lead)
 - Joseph Nwosu
 - Derek Truong
 - Shayaan Chaudhary
 - Ryan Dang
-
- Academic Experience
 - 4 4th year CE undergraduates, 1 3rd year CE undergraduate
 - Digital design using Verilog and FPGA, Cadence, Synopsys
 - Work Experience
 - PCB manufacturing, hardware security research

Digimons

Lead:

Team: EDABKambling

Project proposal

Team: H2T

Lead:

Project proposal

Team: iFSC

Lead:

Project proposal

Inyeon

Team members:

Hifzhi Dinullah <hifzhidinullah>

Ihsan Hidayat Rafi

Nabiel AuFi Danendra

Team background

- Academic Experience

All team members just finished the 3rd year of Electrical Engineering, Institut Teknologi Bandung, Indonesia

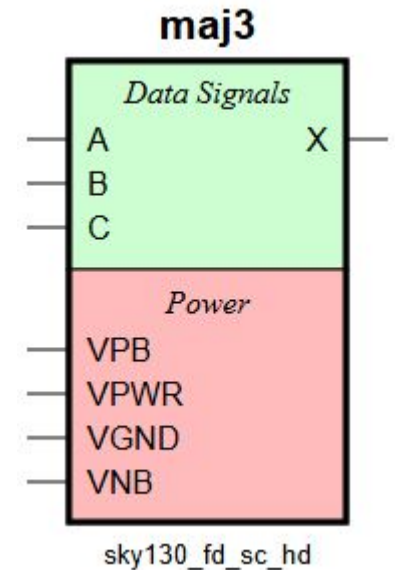
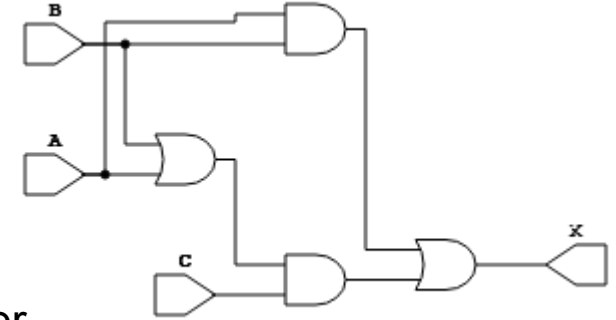
- Work Experience

None

Inyeon

Project information

- Goal:
 - Create a layout of 3 input majority gate with 2x drive strength optimized for power and size efficiency
- Design:
 - Fulfill the logic function of $X = AB + AC + BC$
 - The circuit will be designed so that it have ideal dimension which be easily integrated into the GF180MCU ecosystem
- Application
 - Calculate the carry bit of a full adder
 - Error detection/correction circuit, used in TMR and fault tolerant designs
 - Can acts like a simple logic voting function
- References
 - <https://gf180mcu-pdk.readthedocs.io/en/latest/>



Questions, Suggestions, Doubts?

Lead:

Team: Karnaugh Crew

Project proposal

Mission Silicon

Team members:

Veer Pratap Singh <Whell>

Ahmad Arif Bil Haq

Haoran Du

Rohith Neerati

Shravya Shetty

Team background

- Academic Experience:

Dylan - MEng Biomedical Engineering in Neuroscience

Veer - B.E EEE (3rd year)

Rohith - MS in CE

Arif - Undergraduate Student in Electrical Engineering, Institut Teknologi Sepuluh Nopember

Shravya - B.E EEE (3rd year)

- Work Experience

Dylan - Half-year experience in Mixed-Signal VLSI application

Rohith - None

Veer - Datacenter architecture (documentation)

Mission Silicon

Project information

To extend the existing OSU standard cell library by designing and validating additional logic cells. This will enhance the cell coverage for more efficient logic synthesis.

We propose to design and characterize three new logic cells: XNOR3_1, OR3_1, and NOR3_1, for the 12T track. We aim to extend our contribution as time permits

These additional cells can support improved synthesis and optimization in digital design flows using open-source tools.

References:

<https://gf180mcu-pdk.readthedocs.io/en/latest/index.html>

Questions, Suggestions, Doubts?

Nawiatsob

Team members:

Reynaldo Nicholas <reynaldo1>

Abidah Izdiyar

Mikhael Anacta

Glody Marcello

Team background

- Academic Experience

All members have just finished 2nd Year Electrical Engineering at Institut Teknologi Bandung, Indonesia

- Work Experience

Currently, all members possess no experience at all regarding IC Design

Nawiat sob

Project information

- **Goal:** Designing a new 4:1 multiplexer standard cell with 1X drive strength for the GF180MCU open-source PDK. This cell will expand the existing OSU standard cell library.
- **Design - High Level Proposal:** We're trying to design a 4:1 MUX standard cell using the existing 2:1 MUX standard cell with in the OSU 9-track library,, continuing using the transmission gate configuration to achieve better signal integrity while passing logic, lower power consumption, and more compact area. Our **proposal** is attached below. (<https://docs.google.com/presentation/d/1VRBI1e3FQhSZLru2IltRmNX-giGgi5SduanPqNunnVw/edit?usp=sharing>)
- **Application:** This component is a fundamental building block for data path routing, enabling the selection of one out of four data lines in digital logic circuits such as bus systems, ALUs, and communication processors.
- **References**
https://gf180mcu-pdk.readthedocs.io/en/latest/digital/standard_cells/gf180mcu_fd_sc_mcu9t5v0/cells/mux4/gf180mcu_fd_sc_mcu9t5v0_mux4_1.html

Questions, Suggestions, Doubts?

- We saw couple of other teams planning to design a 4:1 MUX as well, is it possible that multiple teams will be designing the same building block, or should it be discussed between the teams so each team can make different variation (track, drive strength, etc.)?
- We are open to any resources that can help us get a more in-depth understanding of each designing process (schematic, layout, characterization, etc.), given our limited experience in the field.

Lead:

Team: Rhinoceros

Project proposal

Setup & Hold

Team members:

Leader name <fossi-chat name>

Yuri Honegger
Kaviprashanna Krishnamoorthy
Bittu
shrirams

Team background

- **Academic Experience**

Yuri: first year BSc Computer Science (ETH Zurich)
Kaviprashanna Krishnamoorthy: BTech in EEE (NITT)
Bittu: Bachelors in Electronics Engineering
shrirams:

- **Work Experience**

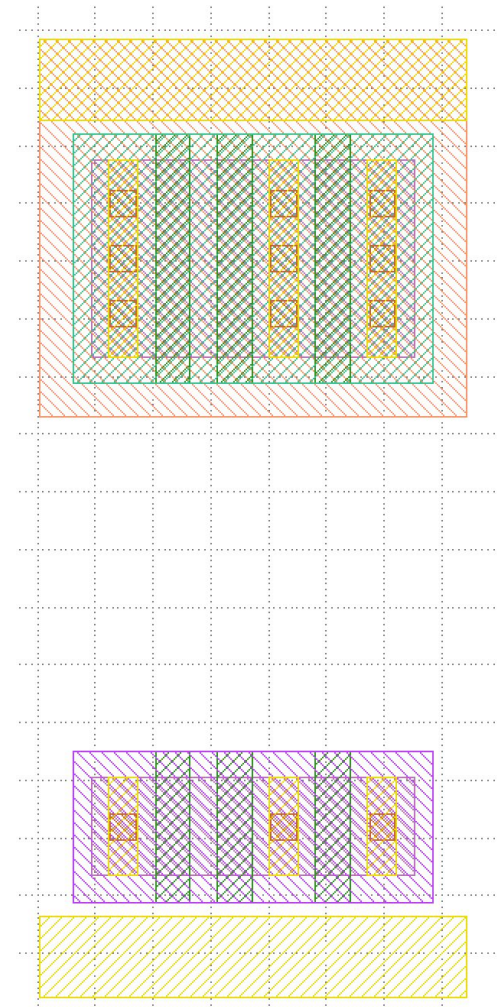
Yuri: none
Kaviprashanna Krishnamoorthy: RTL Engineer (optoML)
Bittu: RTL Engineer
shrirams:

Setup & Hold

Project information

- Goal:
 - Design a MUX4 standard cell for the OSU 12-track library in at least two drive strengths (D1 and D2)
 - Demonstrate improved synthesis results using new cell (only synthesis)
- Design:
 - Not yet finalized, but what is clear:
 - Multiple drive strengths achieved by changing output transistor while the remaining layout stays the same.
 - Transistors generated by klayout standard cell template PCell
- Application: Better synthesis results from having more cells

<https://github.com/diadata/sscs-chipathon-2025-team-setup-and-hold/blob/main/proposal.md>





Team Members:

Aditya Minocha (@adityaminocha:matrix.org) (Lead)
Devansh Raut (@dev_ansh:matrix.org)

Team Background

- **Academic Experience**

1. Aditya Minocha - Final Year EEE, B.Tech Undergraduate Student at Manipal Institute Of Technology
2. Devansh Raut - Final Year EEE, B.Tech Undergraduate Student at Manipal Institute Of Technology

- **Work Experience**

Both the team members have had Internship Experience at FOSSEE, IIT Bombay. The internship involved including Analog and Digital Subcircuits in eSim. Currently, we are looking for opportunities in the semiconductor industry.

Project information

- **Project Goal**

- We aim to design a low-power, area-efficient 6T SRAM bitcell as a custom standard memory cell for the GF180MCU-D process, tailored to open-source design flows. Our objective is to optimize the bitcell for improved PPA (Power, Performance, Area) by enhancing read stability, write margin, and leakage reduction under typical and low-voltage conditions (down to 0.9 V).

- **Design – High-Level Proposal**

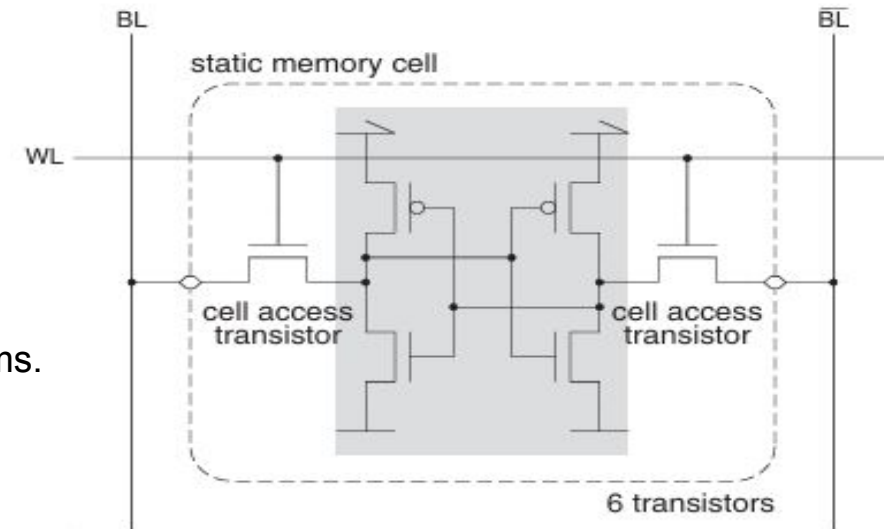
- The 6T SRAM cell will be implemented using the GF180MCU-D PDK, leveraging high-VT devices for improved retention and minimized leakage in standby mode. Our transistor sizing methodology will target balanced SNM and writability, using process-aware simulation sweeps across PVT corners, and improve noise margins.

- **Application**

- This custom memory cell enables integration of reliable, low-power SRAM arrays in open-source SoCs built using GF180MCU, particularly for IoT, and MCU-based systems.

- **References**

- Modica-SRAM (<https://github.com/eda-ricercatore/Modica-SRAM>)
- GF180MCU PDK Docs (<https://gf180mcu-pdk.readthedocs.io/en/latest/>)
- J.P. Kulkarni et al., “A 160 mV Robust Schmitt Trigger Based Subthreshold SRAM,” IEEE JSSC, 2007



Questions, Suggestions, Doubts?

SILICON_BBI1

Lead: <sroutk>

Team members: Saroj Rout <sroutk>, (Lead)
Vishal Kumar Roy, 3rd Yr., ECE
Ananya Satapathy, 3rd Yr., ECE
Aditi Patel, 2nd Yr, ECE
Subhranshu Das, 3rd Yr, ECE

Team background

- Academic Experience
 - Team Leader is the Adjunct professor at Silicon University, India, mentoring both the teams SILICON_BBI1 and SILICON_BBI2.
 - For the rest of the team, 3 are in their 3rd year and 1 in 2nd Yr pursuing undergraduate course in Electronics and Communication Engineering (ECE) at Silicon University, India (<https://silicon.ac.in>)
- Work Experience
 - Team leader has 20+ year experience in analog IC design.
 - Rest of the team have little or no experience in IC design. This summer, few were introduced to basic Verilog and CMOS circuits.

SILICON_BBI1

- **Goal** of this project is twofold: introduce undergraduate students to back-end part of IC design flow using standard cell design and characterization and in the process, contribute to a open source standard cell library (OSU GF180MCU).
- **Design** - Currently there is a matured 5V standard cell library for the open source PDK for GF180MCU process. 3.3V standard cell library developed at Oklahoma State University (OSU) is currently a subset of the 5V standard cell library with basic gates and Flip-Flops with mostly 1X drive strength. Each member will choose a non-existent cell in the library and create the entire library formats for that cell (xschem, GDS, magic, Liberty, LEF, Spice, verilog) and characterize them to make the cell production ready. NOTE: OSU library cells are laidout in gf180mcuC PDK, this project uses gf180mcuD.
- **Application** - of these cells will be used in placing and routing of a digital IP or SoC.
- References :
 - [Global Foundry 5V Standard Cell Library](#)
 - [OSU 3.3V GF Standard Cell Lib](#)
 - [Tim Edwards stdcell suggestions from June 27 Meeting](#)

SILICON_BBI2

Lead

Team members: Prasnjit Biswal <>, 3rd Yr., EEE (Tentative Lead)
Bhabisyajit Bal, 2nd Yr., EEE
Aditya Pratyush Mohapatra, 3rd Yr., ECE
Aditya Sahoo, 2nd Yr, ECE
Sonu Singh, 3rd Yr, ECE
Accepting new members

Team background

- Academic Experience
 - Team consists of mix of 3rd year and 2nd year undergraduate students pursuing Electrical and Electronics Engineering (EEE) or Electronics and Communication Engineering (ECE) at Silicon University, India.
- Work Experience
 - This summer, some of the members were introduced to basic Verilog and the rest to CMOS circuits.
 - This team will work together with SILICON_BBI1 with <sroutk> mentoring this group as well.

SILICON_BBI2

- **Goal** of this project is twofold: introduce undergraduate students to back-end part of IC design flow using standard cell design and characterization and in the process, contribute to a open source standard cell library (OSU GF180MCU).
- **Design** - Currently there is a matured 5V standard cell library for the open source PDK for GF180MCU process. 3.3V standard cell library developed at Oklahoma State University (OSU) is currently a subset of the 5V standard cell library with basic gates and Flip-Flops with mostly 1X drive strength. Each member will choose a non-existent cell in the library and create the entire library formats for that cell (xschem, GDS, magic, Liberty, LEF, Spice, verilog) and characterize them to make the cell production ready. NOTE: OSU library cells are laidout in gf180mcuC PDK, this project uses gf180mcuD.
- **Application** - of these cells will be used in placing and routing of a digital IP or SoC.
- References :
 - [Global Foundry 5V Standard Cell Library](#)
 - [OSU 3.3V GF Standard Cell Lib](#)
 - [Tim Edwards stdcell suggestions from June 27 Meeting](#)

SiliconB

Team members:

- 1) ARPIT BAL <arpitbal> (TEAM LEADER)
- 2) KOWSHIK

Team background

- **Academic Experience:**

- ARPIT- Pursuing PhD from IIT Bombay in Electrical Engineering (Integrated Circuits and Systems)
- KOWSHIK- Pursuing Master's from IIT Bombay in Electrical Engineering (Integrated Circuits and Systems)

- **Work Experience:**

- ARPIT- Researcher at TCS Research(Sep.2020- Dec.2021)
- KOWSHIK - 5G System Engineer at Jio Platform Limited(Jul.2023- Jun.2024)

SiliconB

Project information

- **GOAL:** TO DESIGN 3-INPUT AND 4-INPUT NAND GATES (1X) FOR GF180MCU_OSU_SC_GP9T3V3
- **DESIGN:**

Library	GF180MCU_OSU_SC_GP9T3V3
Cells	1. Gf180mcu_osu_sc_mcu9t3v3__nand3_1 (3 INPUT NAND GATE) 2. Gf180mcu_osu_sc_mcu9t3v3__nand4_1 (4 INPUT NAND GATE)
Drive Strength	1X
Track	9 Track
VDD	3.3 V

SiliconB

DESIGN:

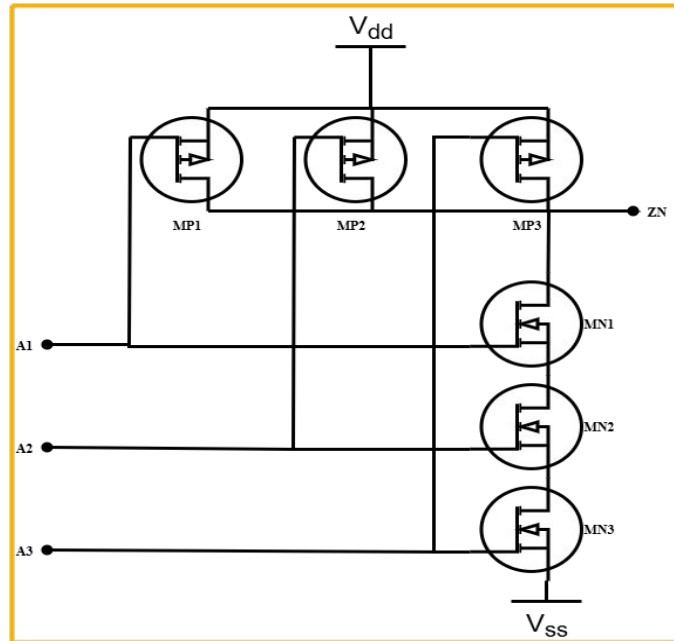


Fig.1 3-INPUT NAND GATE

Parameter	Targeted Spec.	Output Load
Input Capacitance	<0.01 pF	0.001 pF
Leakage Power	<0.15 nW	0.001 pF
Area	<25 μm^2	-
Delay	<0.2 ns	0.001 pF

NAND3_1 TARGETED SPECIFICATIONS

SiliconB

DESIGN:

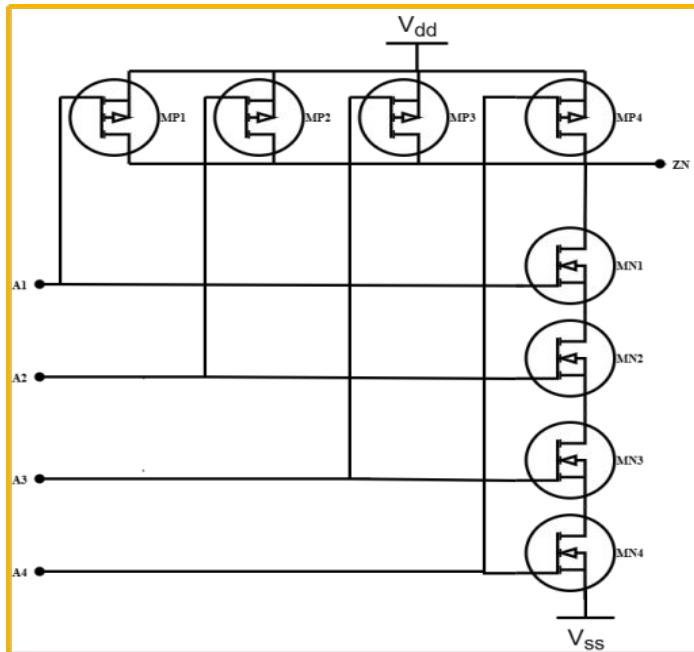


Fig.2 4-INPUT NAND GATE

Parameter	Targeted Spec.	Output Load
Input Capacitance	<0.01 pF	0.001 pF
Leakage Power	<0.2 nW	0.001 pF
Area	<30 μm^2	-
Delay	<0.2 ns	0.001 pF

NAND4_1 TARGETED SPECIFICATIONS

SiliconB

ID	TASK NAME	START	FINISH	DURATION	JUL 2025			AUG 2025				SEP 2025	
					13-7	20-7	27-7	3-8	10-8	17-8	24-8	31-8	7-9
1	DESIGN AND SCHEMATIC LEVEL SIMS	11-07-2025	21-07-2025	7D									
2	LAYOUT+DRC+LVS	22-07-2025	04-08-2025	10D									
3	CHARACTERIZATION	05-08-2025	25-08-2025	15D									
4	INTEGRATION OF CELLS INTO LIBRARY	26-08-2025	08-09-2025	10D									

TENTATIVE TIMELINE

SiliconB

WORK DIVISION

MEMBER NAME	ARPIT BAL	KOWSHIK
BLOCK ALLOTTED	4-INPUT NAND_1x	3-INPUT NAND_1X
DETAILED DESCRIPTION OF TASK ALLOTTED	Schematic+Layout+DRC+LVS+ Characterization of 4-INPUT NAND_1X	Schematic+Layout+DRC+LVS+ Characterization of 3-INPUT NAND_1X

REFERENCES

- <https://gf180mcu-pdk.readthedocs.io/>

Symbelleuit

Team members:

Kean Malik Aji Santoso (@keanmas:matrix.org)

Anas Fathurrahman

Muhammad Yoga Putratama

Team background

- Academic Experience

- Finished a 3rd year in Electrical Engineering, Institut Teknologi Bandung (ITB), Bandung, Indonesia
- Have experience in VLSI design with VHDL and Verilog code;
- Layout and Synthesis from RTL to LVS/DRC checking in SKY130-PDK using open tools via IIC-OSIC-TOOLS.

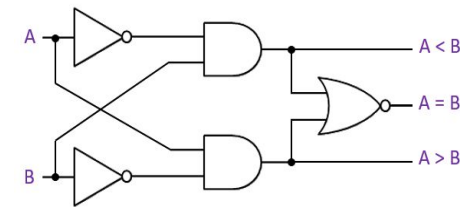
- Work Experience

Currently, we have no work experience regarding digital IC design.

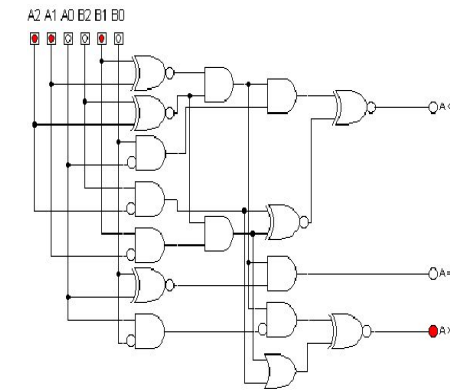
Symbelleuit

Project information

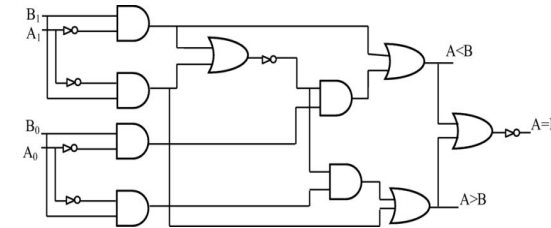
- Goal:
 - We are going to create a 4-bit digital comparator (8-inputs) for the library **GF180MCU_OSU_SC_GP9T3V3** if possible, 3-bit digital comparator (6-inputs) if possible, 2-bit comparator (4-inputs) digital comparator if possible, or minimum 1-bit digital comparator (2-inputs) for the same library because other PDK, such as **SKY130-PDK**, already has a standard cell of comparator.
- Design:
 - Digital comparators
 - 1-bit digital comparator (gf180mcu_osu_sc_gp9t3v3__comp2_x)
 - 2-bit digital comparator (gf180mcu_osu_sc_gp9t3v3__comp4_x)
 - 3-bit digital comparator (gf180mcu_osu_sc_gp9t3v3__comp6_x)
 - 4-bit digital comparator (gf180mcu_osu_sc_gp9t3v3__comp8_x)
 - Drive strength is 1x. If possible, we would try to implement a drive strength of 2x.
- Application: CPU & microcontrollers for “if-else” statements, address decoding, ADC
- References
 - [Design of a 4-bit comparator](#)
 - [Magnitude Comparator and Digital Comparator : Types & Their Applications](#)
 - [» Magnitude Comparator - World Of Computing](#)
 - [SkyWater Foundry Provided Standard Cell Libraries](#)
 - [9 track Standard Cells — GlobalFoundries GF180MCU PDK 0.0.0-111-gde3240d documentation](#)
 - [A New Nano Design for Implementation of a Digital Comparator Based on Quantum-Dot Cellular Automata](#)
 - https://www.researchgate.net/publication/228664794_Evolutionary_Design_of_Digital_Circuits_Using_Improved_Multi_Expression_Programming_IMEP
- Detailed Proposal: [SSCS-Chipathon-2025_Project-Proposal_Symbelleuit](#)



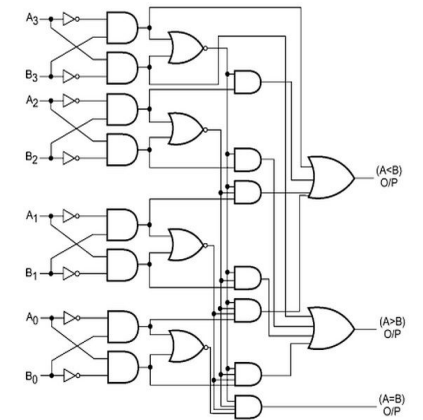
gf180mcu_osu_sc_gp9t3v3__comp2_x



gf180mcu_osu_sc_gp9t3v3__comp6_x



gf180mcu_osu_sc_gp9t3v3__comp4_x



gf180mcu_osu_sc_gp9t3v3__comp8_x

Lead:

Team: SystemsGenesys

Project proposal

Team: TV

Lead:

Project proposal

Lead:

Team: The Short Circuits

Project proposal

VoidWalkers - Scan DFF for DFT

Team members:

Ahmad Jabar Ilmi <Aji Ilmi> [Team Leader]

Aakarsh A

Open for 1 more members! Contact us on Elements.

Team background

- **Academic Experience:**

- AJI - Final year Electrical Engineering Undergraduate Student at ITS
- Aakarsh A - Final year Electrical Engineering Undergraduate Student at IITK

- **Work Experience:**

- AJI - Universalization of IC Design CASS (IEEE) 2024 Participant
- Aakarsh A -

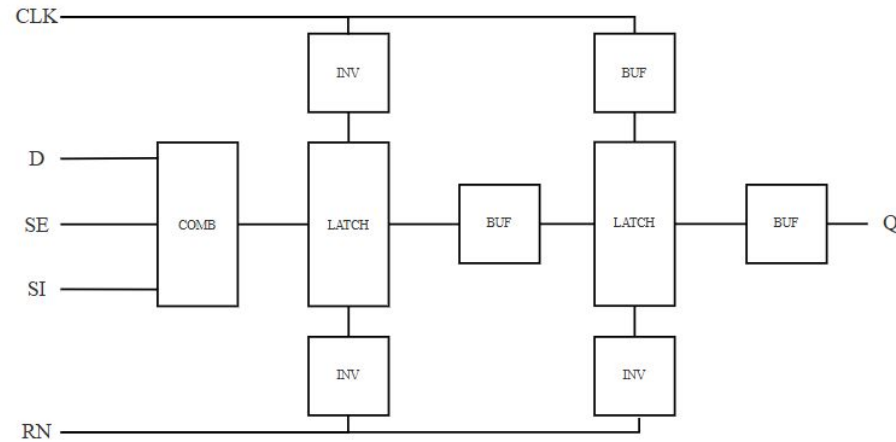
VoidWalkers - Scan DFF for DFT

- **GOAL:** To design positive edge triggered D-type flip flop with active low reset and scan capability for GF180MCU OSU PDK
- **DETAILS:**

Library	gf180mcu_osu_sc_gp9t3v3__sdffrnq_1
Drive Strength	at least 1X, preferably stronger
Track	9 Track
Supply	3.3 V
Ports	IN: CLK, D, SI, SE, RN OUT : Q
Proposed Fmax	at least 100 MHz (timing parameters not yet known), preferably higher or close to gf180mcu_fd_sc_mcu9t5v0__sdffrnq_1

VoidWalkers - Scan DFF for DFT

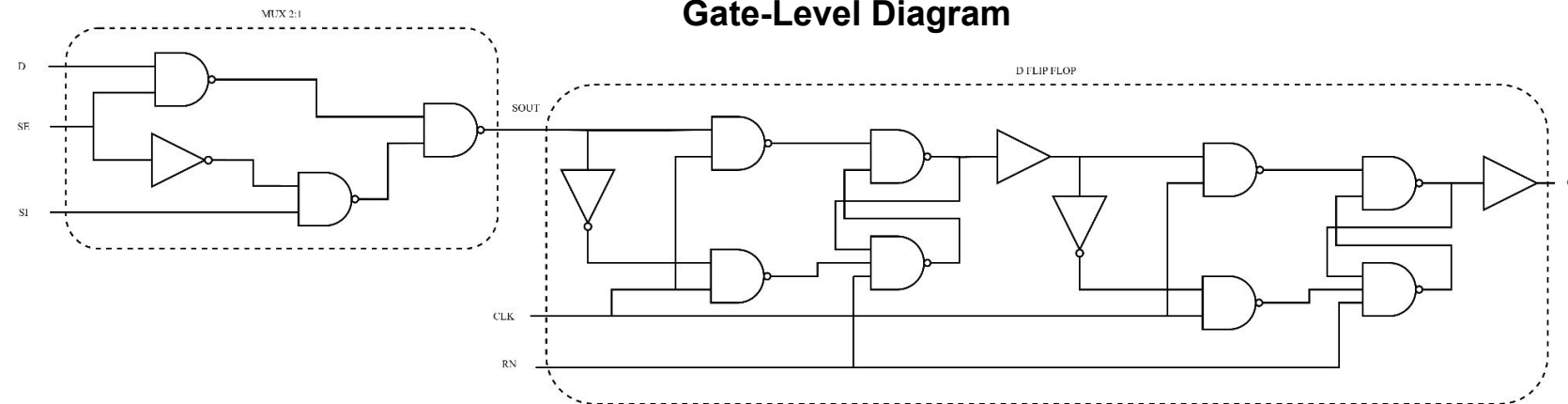
Functional Block Diagram



For more info, please visit our repo at :

[radityankn/voidwalkers-scandff-gf180mcu](https://github.com/radityankn/voidwalkers-scandff-gf180mcu)

Gate-Level Diagram



MOSbius Track Project Reviews (10)

- Chip Odyssey
- CreActive
- DAC to the Future
- FPTA
- Low Noise Anomaly
- Nirvana
- Prof Morbius
- RF_MOSbius
- SpikCore
- XDAC

AC3E Chile

Team members:

jorge_marin

Team member name 1

Team member name 2

Team member name 3

Team member name 3

Team member name 4

Team member name 5

Team background

- Academic Experience
- Work Experience

AC3E Chile

Project information

- Goal (2 sentences)
- Design - High Level Proposal (2-3 sentences + Image and/or link to details)
- Application (1 sentence)
- References

Questions, Suggestions, Doubts?

-

Lead:

Team: Chip Chip Hooray

Project proposal

Lead:

Team: Chip Odyssey

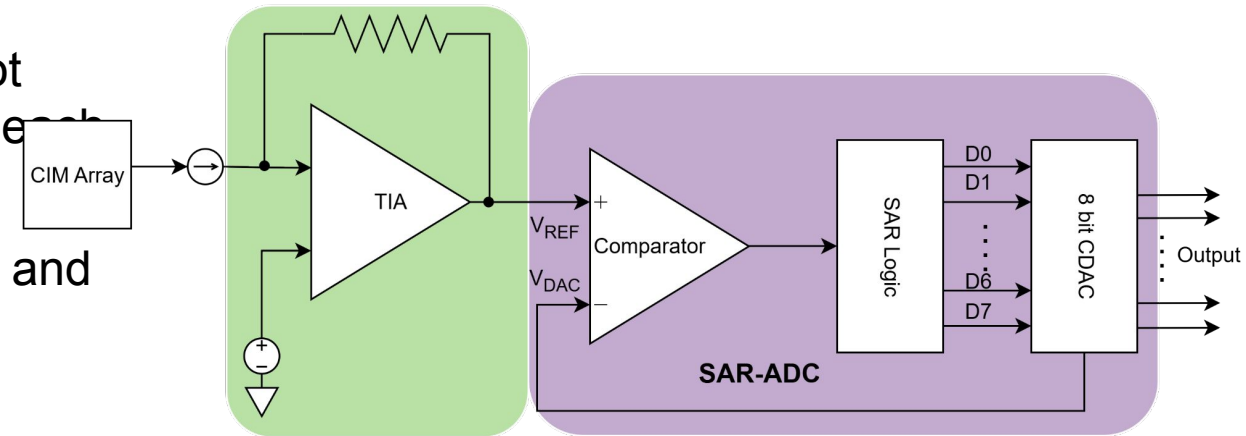
Project proposal: Readout IC for Compute-in-Memory (ROIC-CIM)

Team Information and experience: [Name (fossi) - background(experience)]

- | | | |
|--------------------------------|---|---|
| Mia Kadam (meghanakadam) | - | PhD researcher, (Custom IC EDA, no open source tools) |
| Fabian Schenzle (fab1an) | - | Master Student, (Custom IC EDA, no open source tools) |
| Emmanuel (insightful_sherlock) | - | Master Graduate, (Custom IC EDA, open source tools) |
| Kevin (kejor) | - | Master Student, (Custom IC EDA, open source tools) |
| Rahul Sharma (rahul.shsr) | - | Post Doc researcher, (Custom IC EDA, open source tools) |
| Nitin Indukuri (nindukuri) | - | Master Student, (Custom IC EDA, open source tools) |

Team: Chip Odyssey

- **Goal:** We propose the design of basic concept of a dot product through analog computing and demonstrates each building blocks to the MOSbius users.
- **Design:** The proposed system comprises an OPAMP, and a SAR-ADC containing a comparator, SAR Logic, a capacitive DAC and output buffers
- **Application:** In-memory computing readout circuit, highly parallel Vector-matrix multiplication
- **References:** A. Firdauzi et al., "Power Efficient Current-Mode SAR ADC for Memristor Readout in 28 nm CMOS," 2024 31st IEEE International Conference on Electronics, Circuits and Systems (ICECS), doi: 10.1109/ICECS61496.2024.10848856.



Doubts/ Question?

- Switch Matrix
- Area allocation

Lead:

Team: Chipa Chapa

Project proposal

CreActive

Team members:

Nahwan Faza Assaify <Nahwan Assaify>

Yohanes Stefanus

Open for 3 more members! Contact us on element.

Team background

- Academic Experience

Nahwan Assaify - Final-year Electrical Engineering Undergraduate Student

Yohanes Stefanus - Final-year Electrical Engineering Undergraduate Student

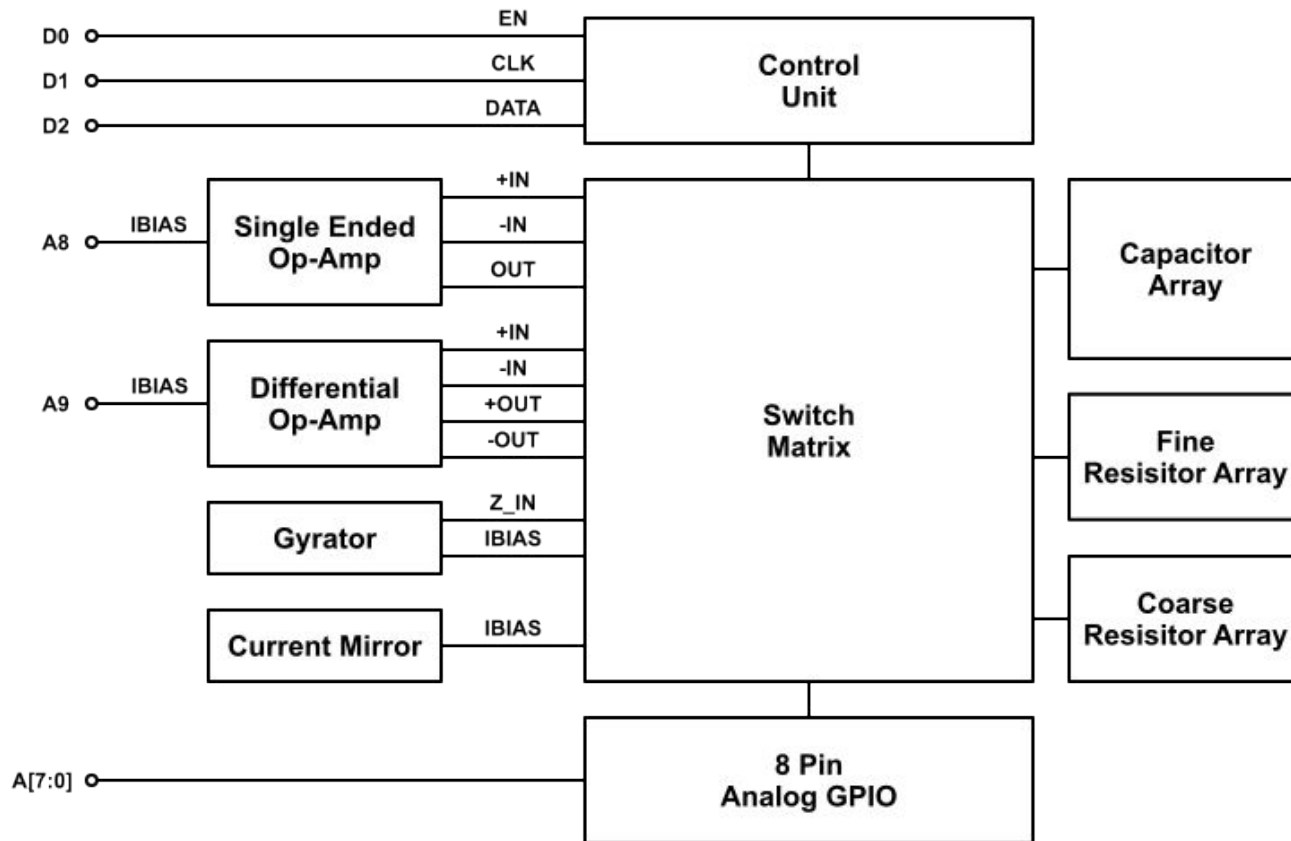
- Project Experience

Nahwan Assaify - Universalization of IC Design CASS (IEEE) 2024 Participant

Yohanes Stefanus - Analog & Mixed-Signal IC Design Research Intern at NYCU, Taiwan

CreActive

Project information



- Goal: explore the possibility of translating filter transfer functions into an analog circuit
- GitHub Repository:
<https://github.com/assaify/creactive-chipathon-2025>
- Application: variety of active filter topologies; sallen-key, state-variable, biquad etc.
- References
 - [1] D. Lancaster and D. Lancaster, Don Lancaster's active-filter cookbook /, 2nd ed. Oxford : Newnes, 1996.
 - [2] H. Guo, Y. Zhang, X. Tang, J. Li, F. Huang, dan N. Jiang, "Design of CMOS Bandwidth Tunable Low-Pass Filter," 2020 5th Int. Conf. Integr. Circuits Microsystems, ICICM 2020, vol. 2, no. 4, hal. 302–305, 2020, doi: 10.1109/ICICM50929.2020.9292272.
 - [3] B. Razavi, "The Design of a Biquadratic Filter [The Analog Mind]," in IEEE Solid-State Circuits Magazine, vol. 16, no. 1, pp. 6-13, winter 2024, doi: 10.1109/MSSC.2023.3336149.

DAC to the Future

Team members:

Kevin Kong <Kevin Kong>
Amirmohammad Fathalian
Jimena Martinez Cano
Jatin
Andrea Sannino

Team background

- Experience

Jimena Martinez Cano - M. Sc. Student in Microelectronics (Currently employed as Analog-IC designer)

Amir Mohammad Fathalian - M.Sc. Student in Embedded Computing Systems - Microsystems

Jatin - Masters in Electronics Engineering (Analog/RF design Engineer)

Kevin Kong - Pursuing PhD in Electrical Engineering (Currently lecturer)

Andrea Sannino - Pursuing PhD in Electronics Engineering

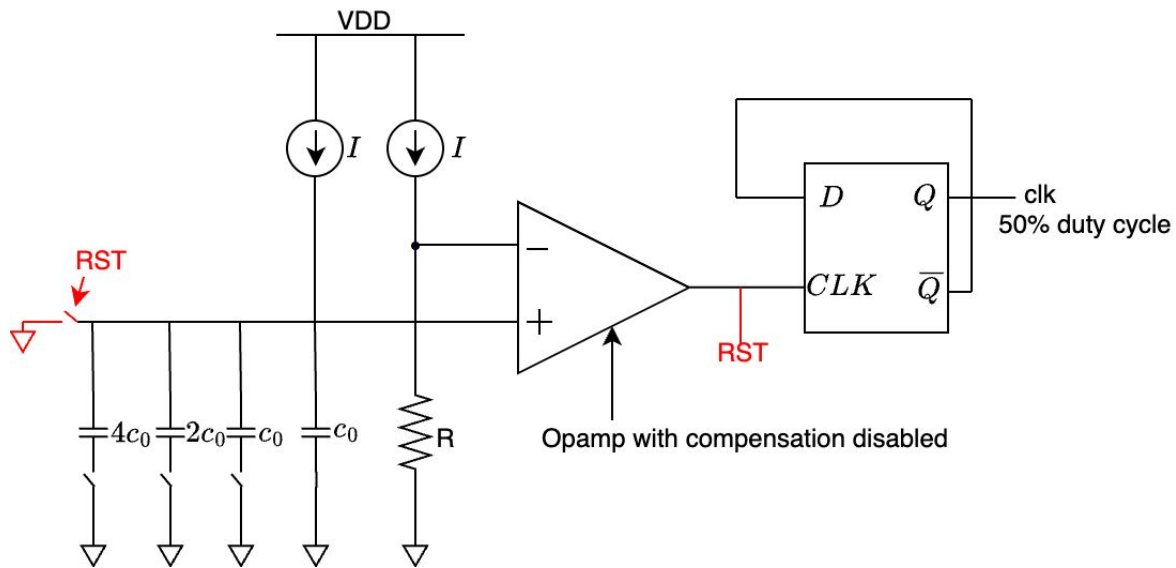
DAC to the Future

Project information

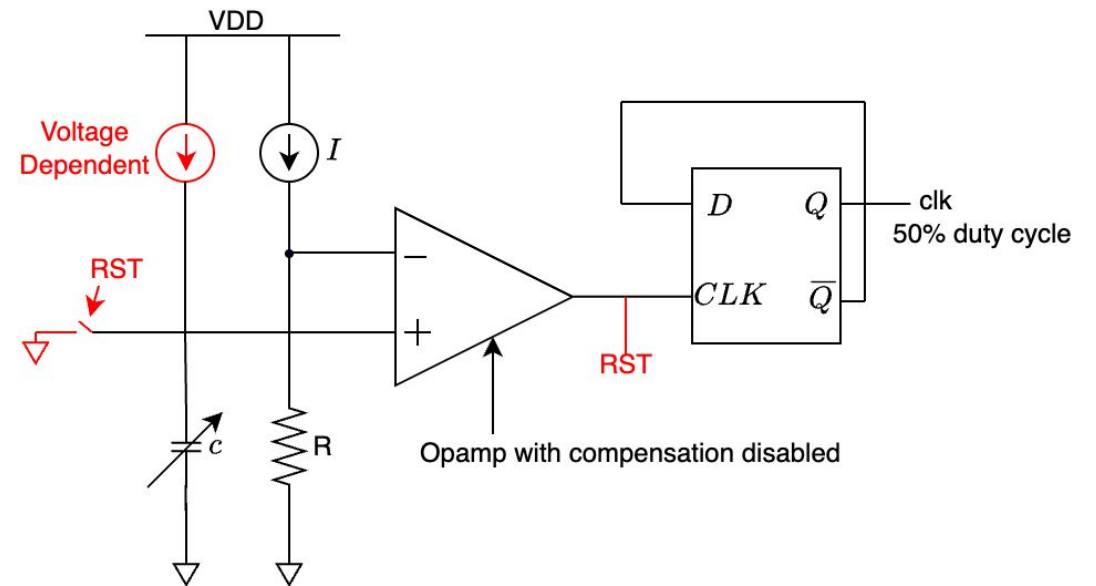
- Implement DAC, LDO, 2 Stage OTA, Programmable VCO, Programmable RC oscillators on MOSbius
Each block can be studied on its own, including the reference generators (bandgap/constant G_m)
- Design
 - Start by designing a 2 Stage OTA
 - Use the OTA for different blocks, as stated above
- Application
 - Educational purposes for advanced analog design course
 - Learn and test OTA applications: DAC, LDO, Bandgap, VCO, etc.

M0	M1	Functionality
0	0	Programmable RC Fixed Frequency Oscillator
0	1	Voltage control oscillator
1	0	LDO
1	1	3 bit DAC

Mode = 00

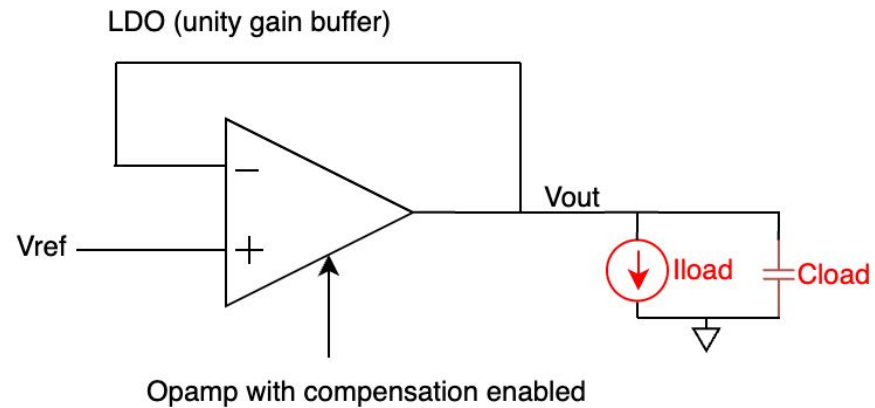


Mode = 01

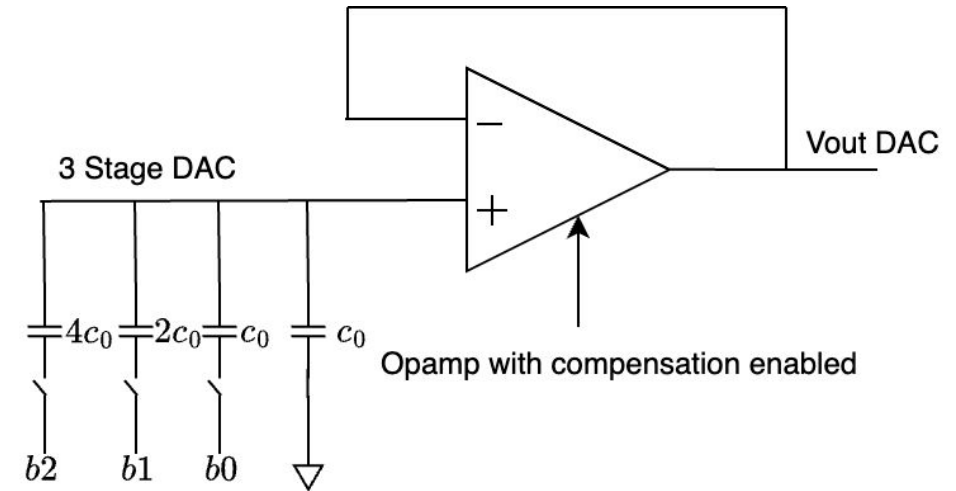


Lead:

Mode = 10



Mode = 11



Schedule

Week 28-33	Schematics and Simulation
Week 34-39	Layout and Verification

Questions, Suggestions, Doubts?

●

FPTA

Team Members: Jason Lyst (Team Lead)
Sanithu Heengama
Ushan Fernando
Accepting new members

Team Background:

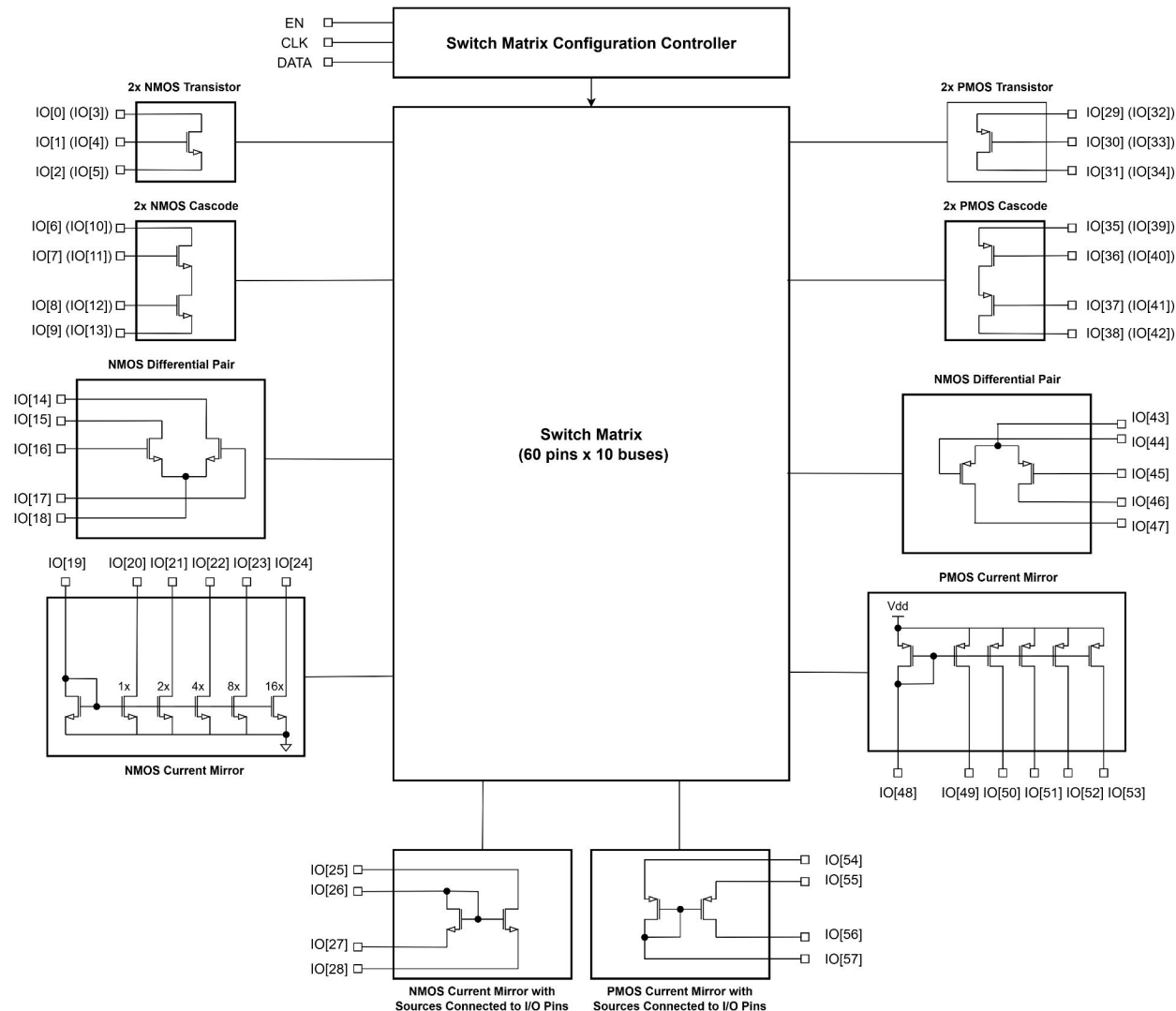
- Academic Experience:
 - Jason Lyst- ECE undergrad 3rd year at Purdue University.
 - Sanithu Heengama- Electrical Engineering undergrad 2nd year at Brown University.
 - Ushan Fernando- Electrical and Electronic Engineering 4th year undergrad at University of Nottingham Malaysia.

FPTA

Project Information:

- **Title:** Design of Amplifier Circuits using the MOSbius Architecture
- **Goal:** To support hands-on learning and deepen intuition in analog circuit design for electrical engineering students and hobbyists learning analog IC design, we propose a MOSbius-based Field Programmable Transistor Array (FPTA) architecture that enables reconfiguration and testing of multiple amplifier topologies.
- **Application:** By dynamically switching transistor-level connections, users can explore and compare different analog amplifier designs from simple designs such as common source and cascode amplifiers to designs such as 5-transistor OTAs, 2-stage OTAs, and telescopic cascode OTAs to name a few.
- **Proposal Link (for more details):**
<https://github.com/ushan-fernando/SSCS-Chipathon-2025-Team-FPTA/blob/proposal/docs/proposal/proposal.md>

FPTA



Tentative Design Block Diagram

Questions, Suggestions, Doubts

- How do we effectively model the parasitics caused by the switch matrix when interconnecting the different blocks to make amplifiers?
- Do we have to account for driving/configuring the GPIO pins?
- Do we also have to design circuits for consistent voltage reference and voltage regulation for power supply?
- Is the scope of our proposal too big for the time-frame? Our team members are all beginners with no prior experience in analog IC design.

Team: iFSC

Lead:

Project proposal

Low Noise Anomaly

Team members:

M Taufiqul Huda <M Taufiqul Huda> [Team Leader]

Raditya Eka Putra

Elijah Teti Johnson

Dzaki Ardiansyah

Adam Ghieh

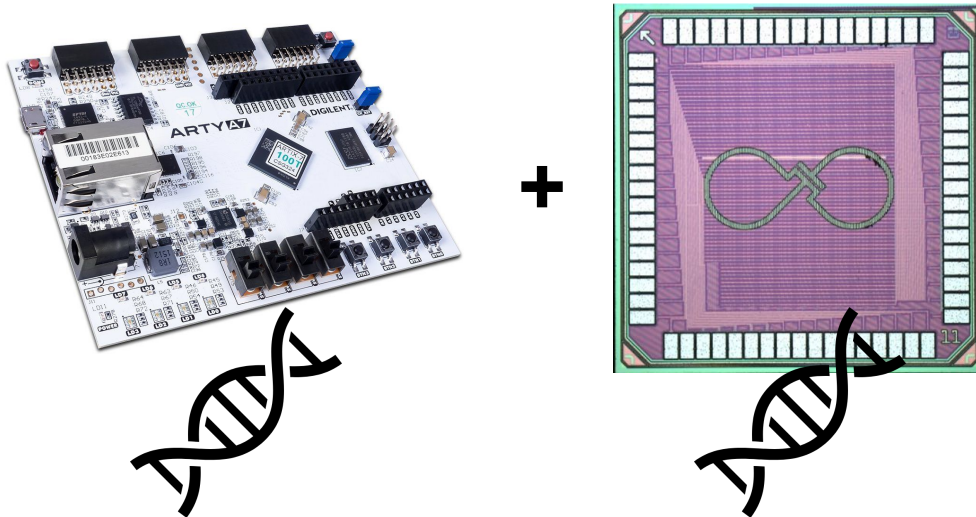
Team background

- Academic Experience
 - M Taufiqul Huda, Raditya Eka Putra & Dzaki Ardiansyah: Undergraduate Student in Electrical Engineering, Institut Teknologi Sepuluh Nopember
 - Elijah Teti Johnson : Undergraduate Electrical Engineering at Columbia University
 - Adam Ghieh : Undergraduate Electrical Engineering at Concordia University
- Work Experience
 - M Taufiqul Huda : Analog IC Lab Intern at NTUST & Chipalooza Participant 2024
 - Raditya Eka Putra : Universalization of IC Design from CASS (IEEE) Participant 2024

Low Noise Anomaly

Project Background and Goal

- The MOSbius architecture integrates analog and digital circuits for transistor-level learning and direct simulation from schematic designs.
- It utilizes a Field Programmable Mixed-Signal Array (FPMA) to emulate analog and digital blocks, offering flexibility for various circuit configurations.



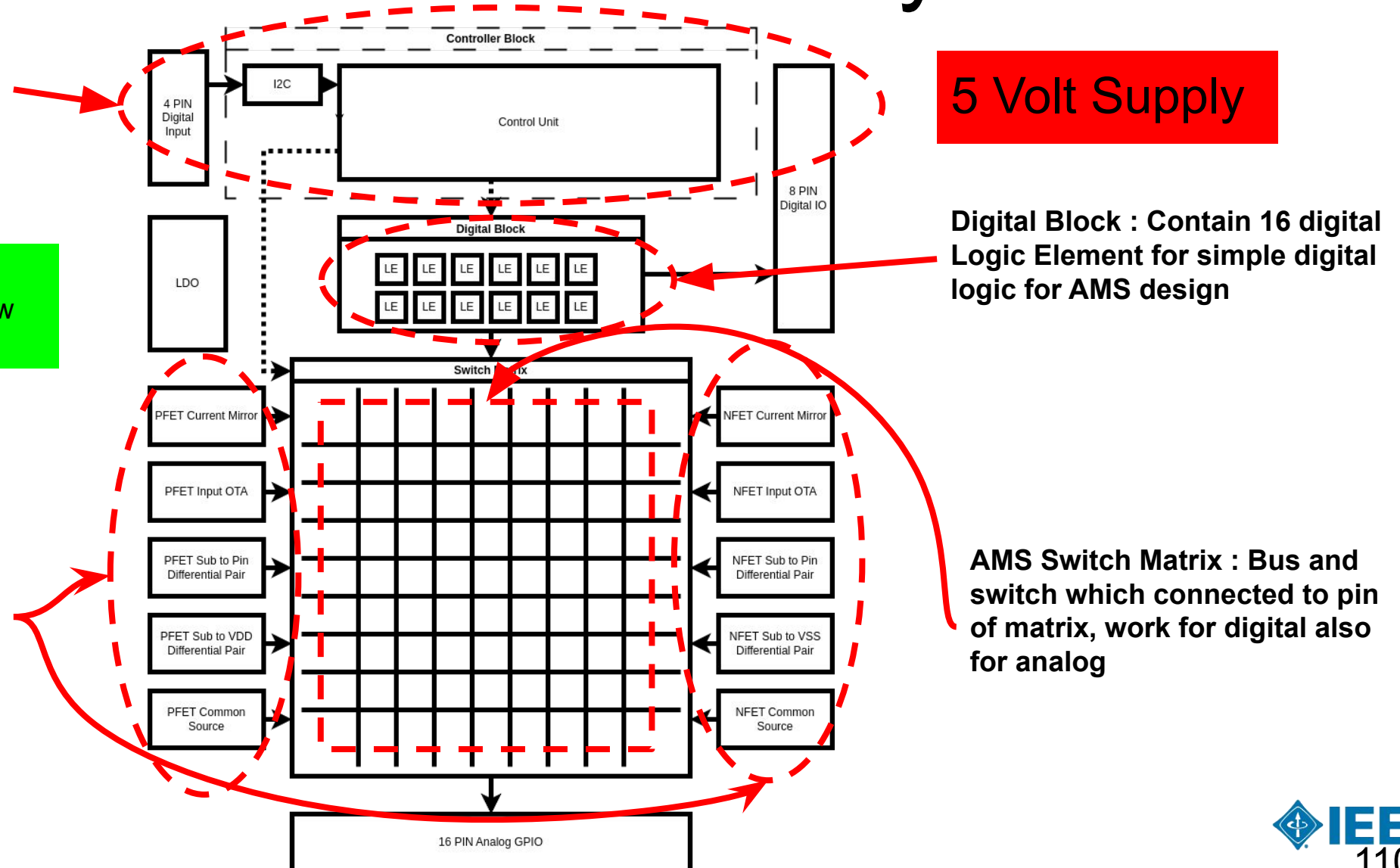
MIXED-SIGNAL MOSBIUS

Low Noise Anomaly

Controller Block : For control the switch matrix register and Logic Element register also programmable using I2C communication

Design Method :
Analog >>> Custom Analog Flow
Digital >>> Digital Flow

Analog Block : Contain lot of individual FET for flexibility of analog design also contain basic block of current mirror and OTA.

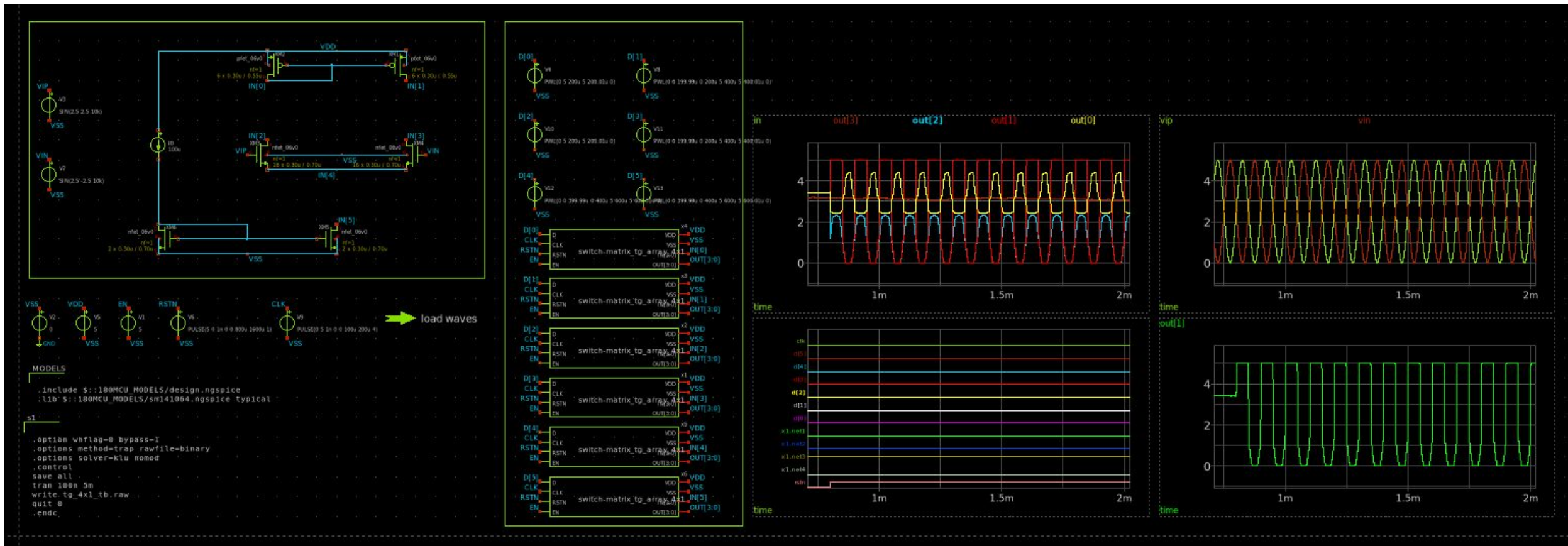


5 Volt Supply

Digital Block : Contain 16 digital Logic Element for simple digital logic for AMS design

AMS Switch Matrix : Bus and switch which connected to pin of matrix, work for digital also for analog

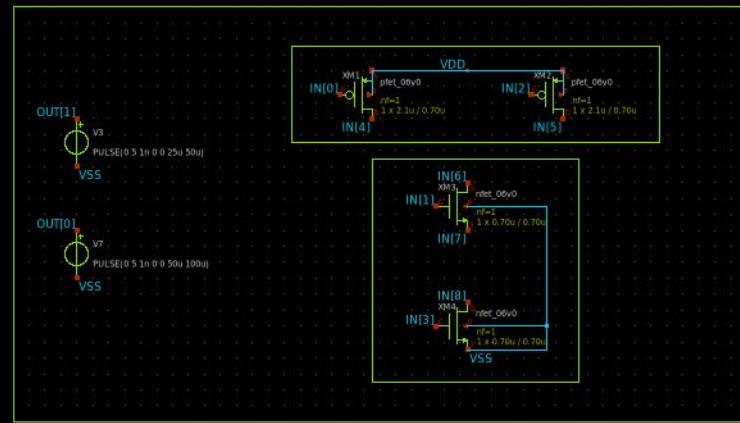
Low Noise Anomaly



5T-OTA Mock Testbench Connected to Switch Matrix

Low Noise Anomaly

2-NAND



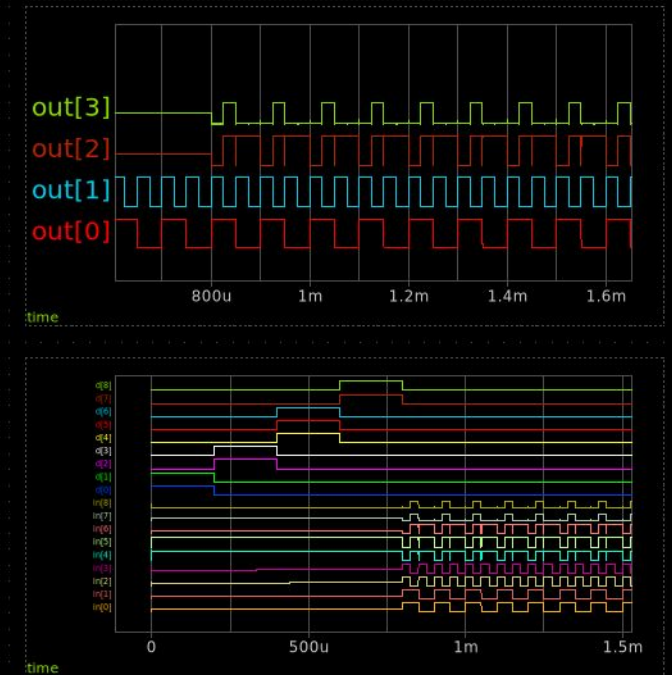
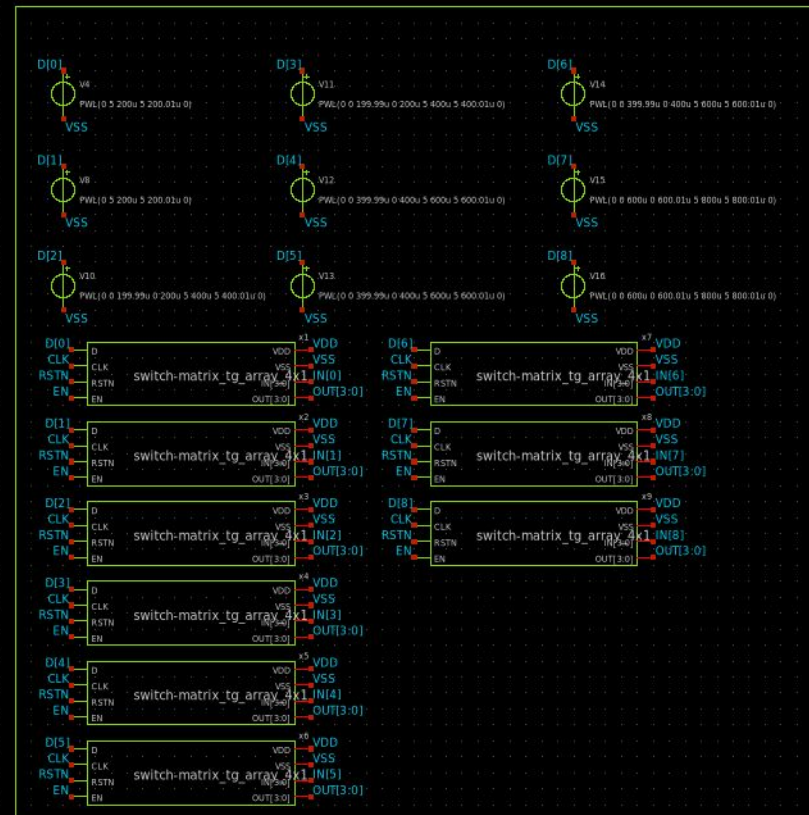
MODELS

```
.include $:/180MCU_MODELS/design.ngspice
.lib $:/180MCU_MODELS/sm141064.ngspice typical

s1
.option wnlag=0 bypass=1
.options method=trap rawfile=binary
.options solver=klu nomod
.control
save all
tran 100n 2m
write tb_switch-matrix_tg_array_4x9.raw
quit 0
.endc
```



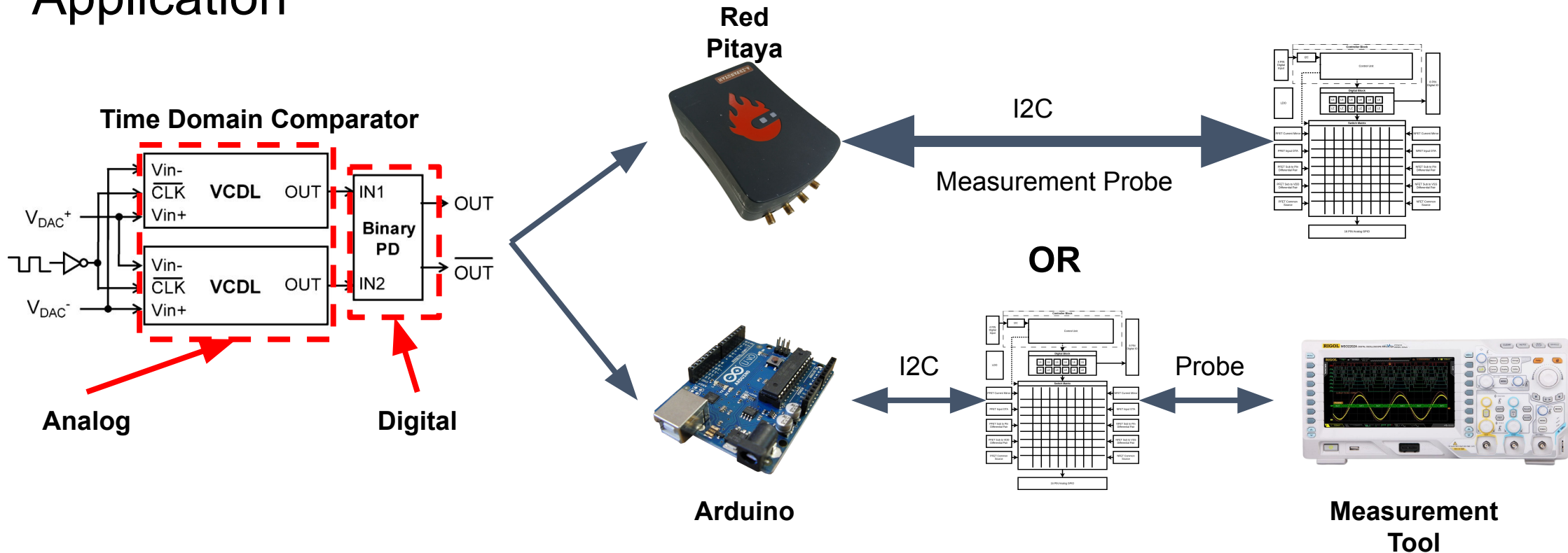
SWITCH MATRIX AND 4 BUS



2-NAND Mock Testbench Connected to Switch Matrix

Low Noise Anomaly

Application



Reference

[1] P. Chow, P. G. Gulak and P. Chow, "A Field-Programmable Mixed-Analog-Digital Array," *Third International ACM Symposium on Field-Programmable Gate Arrays*, Napa Valley, CA, USA, 1995, pp. 104-109, doi: 10.1109/FPGA.1995.242048.

Project Repository

<https://github.com/mthudaa/sscs-chipathon2025-LNA>

Lead:

Team: Lumut Bius

Project proposal

Lead:

Team: mas pri

Project proposal

Lead:

Team: MOSbius Team

Project proposal

Nirvana

Lead: Faiz Setya K.

Team members:

Faiz Setya K. <faizzsk.> [Team Leader]

Nithin P

Kelton Eckert

Kandias Wardhana

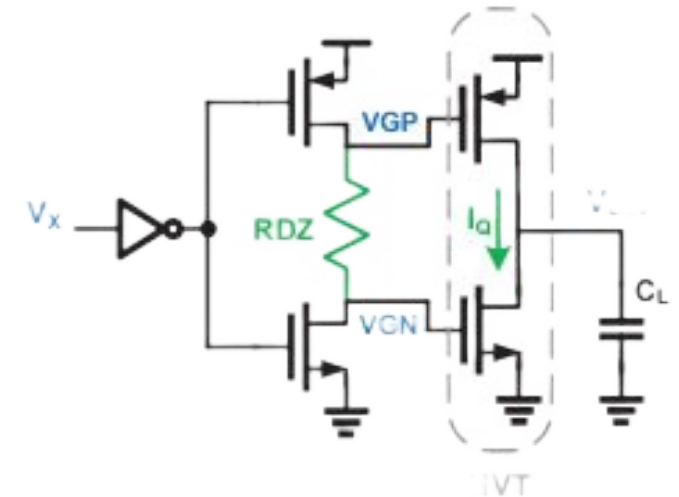
Amit Kumar Sharma

Team background

- Academic Experience
 - Faiz Setya K. & Kandias Wardhana: Undergraduate Student in Electrical Engineering, Institut Teknologi Sepuluh Nopember
 - Nithin P : B.Tech in Electronics and Communication Engineering
 - Kelton Eckert : Undergraduate Electrical Engineering at Brown University
 - Amit Kumar Sharma :
- Work Experience
 - Nithin P : Researcher at Indian Institute of Technology (IIT) Gandhinagar.

Project Background and Goal

- Dynamic amplifiers, particularly Ring Amplifiers(RAMP) was originally proposed by Benjamin Hershberg at Oregon State University (OSU), ring amplifiers offer an innovative solution for high-performance analog blocks, especially in pipelined ADCs and filter circuits.
- Our project aims to design and develop a **Reconfigurable Ring amplifier** optimized for **educational use** and **mixed-signal system integration**. The central focus is on enabling user-driven exploration and understanding of dynamic amplifier behavior—including large-signal dynamics, deadzone effects, and the transition into small-signal operation.

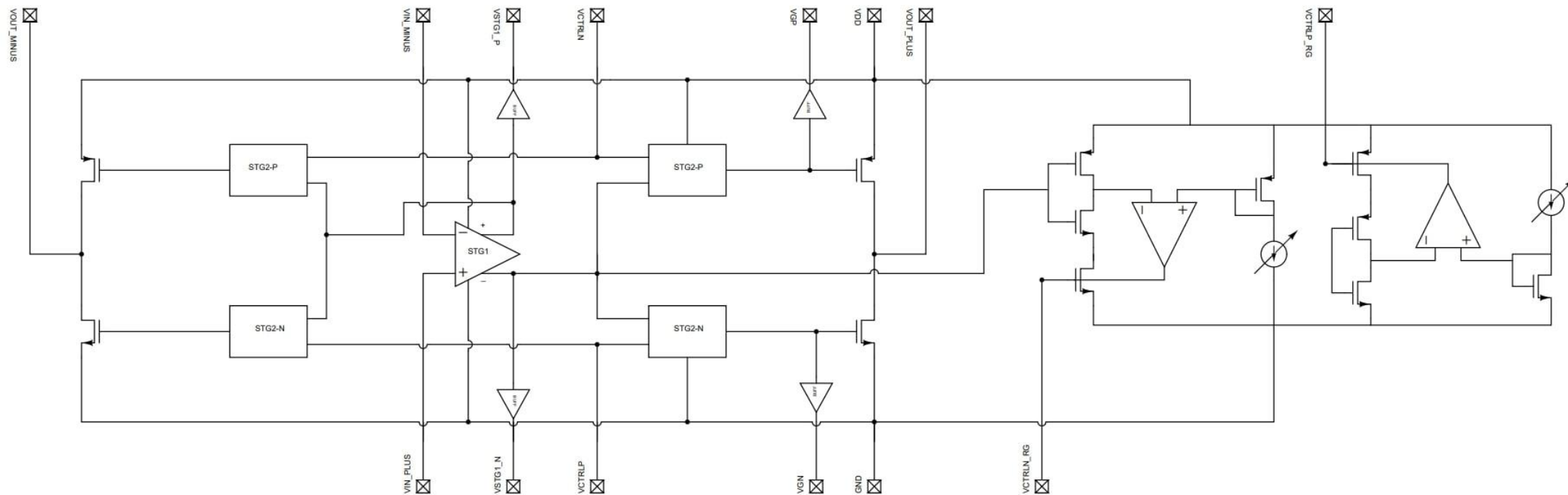


Ring Amp Architecture

Nirvana

CHIPATHON-MOSBIUS TEAM NIRVANA

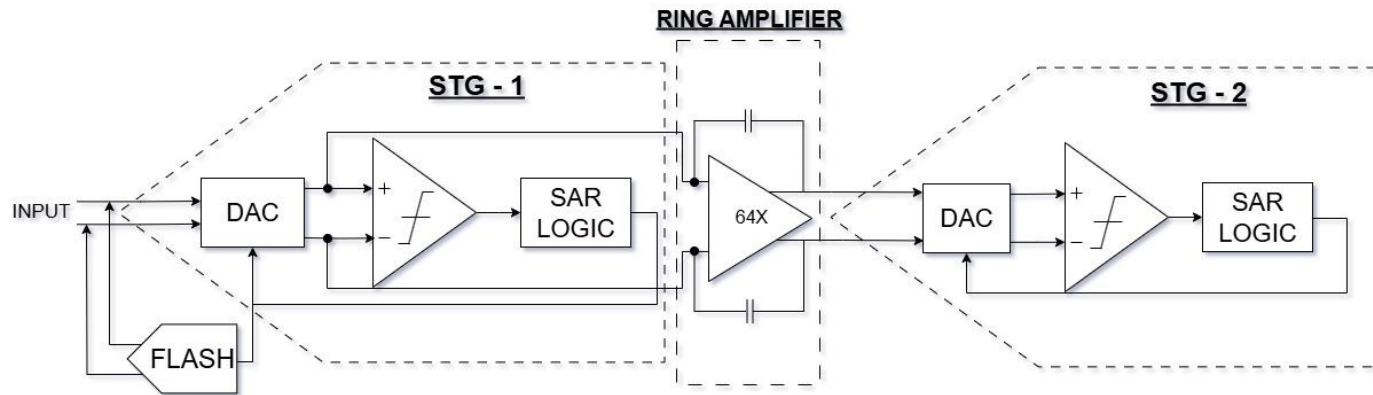
Team Members:
 1. Nithin P. (Design and Architecture)
 2. Kaiton Eckert (Design)
 3. Kandas Warchana (Layout)
 4. Faizz (Layout)
 5. Snigdha Sarai (Layout)



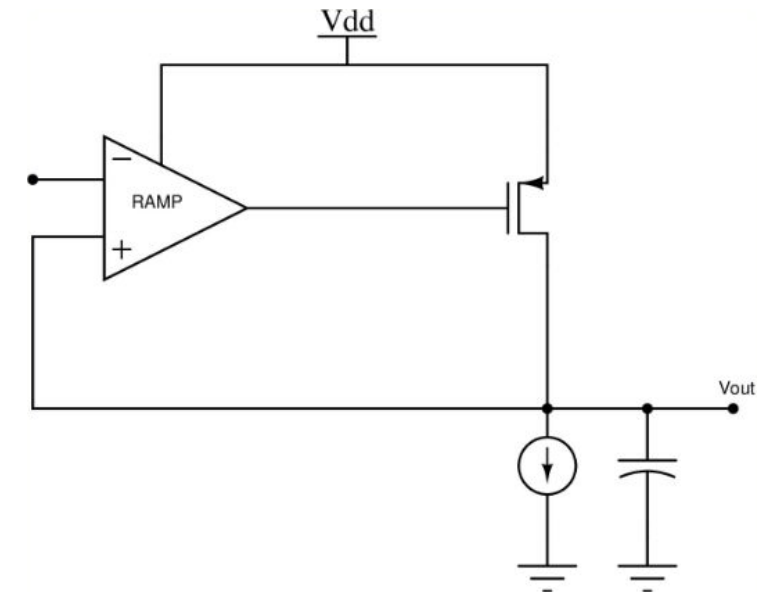
Nirvana

Lead: Faiz Setya K.

Application



Two-Step SAR ADCs



Ring Amplifier Based LDO

Reference

- [1] A. ElShater et al., "A 10-mW 16-b 15-MS/s Two-Step SAR ADC With 95-dB DR Using Dual-Deadzone Ring Amplifier," in IEEE Journal of Solid-State Circuits, vol. 54, no. 12, pp. 3410-3420, Dec. 2019, doi: 10.1109/JSSC.2019.2943935.
- [2] P. K. Venkatachala et al., "Process Invariant Biasing of Ring Amplifiers Using Deadzone Regulation Circuit," 2018 IEEE International Symposium on Circuits and Systems (ISCAS), Florence, Italy, 2018, pp. 1-5, doi: 10.1109/ISCAS.2018.8351242.

Project Repository

<https://github.com/chennakeshavadasa/SSCS-Chipathon-2025-Team-Nirvana>

Lead:

Team: On Chip Team

Project proposal

Prof Morbius

Team members:

James Patrick <[@orpheus016:matrix.org](mailto:@orpheus016@matrix.org)> [Lead]

Ibrahim Hanif

Goldwin Sonick

Team background

- Academic Experience

- We're 3rd year undergraduate engineering student from ITB (Bandung Institute of Technology, ongoing to be the 4th year)
- We have finished the Integrated Circuit Design Analysis Course where we designed a MUX_DFF Layout Design using the IIC-OSIC-TOOLS (SKY130 PDK, Magic, Xschem, Ngspice, Netgen (DRC-LVS), etc).
- For the final project, we designed a complete chip design for a digital LMS Filter implementation using OpenLane2 and Verilog for the RTL.

Prof Morbius

Project information

- **Goals**
 - Design a (Frequency Hopping Spread Spectrum) FHSS Educational Learning Kit with Custom Analog Chip and Comprehensive Learning Guide.
 - Design a custom IC with core analog blocks of a frequency synthesizer (VCO, PLL) with system integration to the off-chip components (on MOSbius Board)
- **Design**
 - On-Chip (Custom IC), we will design 1) **Voltage-Controlled Oscillator (VCO)** to generate the carrier and 2) **Phase-Locked Loop (PLL)** components: Phase Detector, Charge Pump, and Loop Filter to ensure frequency stability and control.
 - We will use the MOSbius board to make 1) **PN Sequence Generator** to control the frequency hopping pattern, 2) **Mixer** for receiver de-hopping and signal demodulation 3) **Modulator** to impress data onto the hopping carrier 4) **Stable Reference Oscillator** for the PLL.
- **Application**
 - Education and Experimentation: A hands-on platform for teaching students system-level principles of spread spectrum communication.
- **References**
 - Example of DSSS/FHSS Trainer Kit: <https://www.tescaglobal.com/product/cdma-dsss-fhss-trainer>

Prof Morbius

Questions, Suggestions, Doubts?

- We're new to RF in general, (we only learnt the basics in analog & digital modulation/demodulation) do you think it's feasible for us to learn and understand this topic and then implement it in this chipathon? If it isn't should we change tracks or just make a simpler project? Please give us your suggestions.
- This idea stems from our interest on measures to avoid interferences that we can implement on a chip-level. An alternative we found was DSSS (Direct-Sequence Spread Spectrum) but we haven't really found a good source about this. Any recommended references is helpful.
- Should the mixer be made using the MOSbius chip, or just a COTS one integrated into the PCB testing kit? We haven't really finalized our system structure.
- Is it better if we include more applications for this chip or focus on spread spectrum communication?

Lead:

Team: RAZAVUS

Project proposal

RF_MOSbius

Team members:

Andrew Chon <andrewchon>

Xianglin Pu

Qizhe Wu

Jingde He

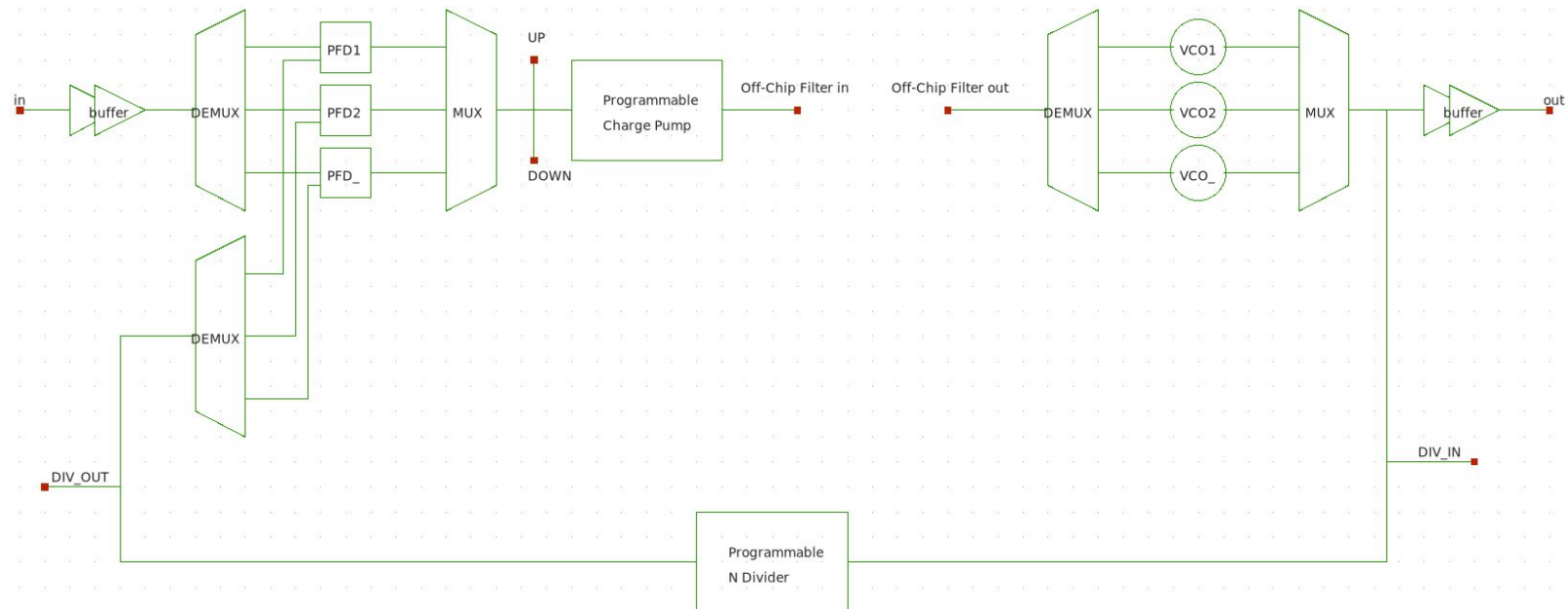
Team background

- Graduate Research Students of the Columbia Integrated Systems Lab, Analog and RF IC Research Group at Columbia University in the City of New York.
- We have tape-out experience using commercial tools (Cadence) in a different process (TSMC). However, we are new to the Open Source tools and the GF180 process used in this chipathon.

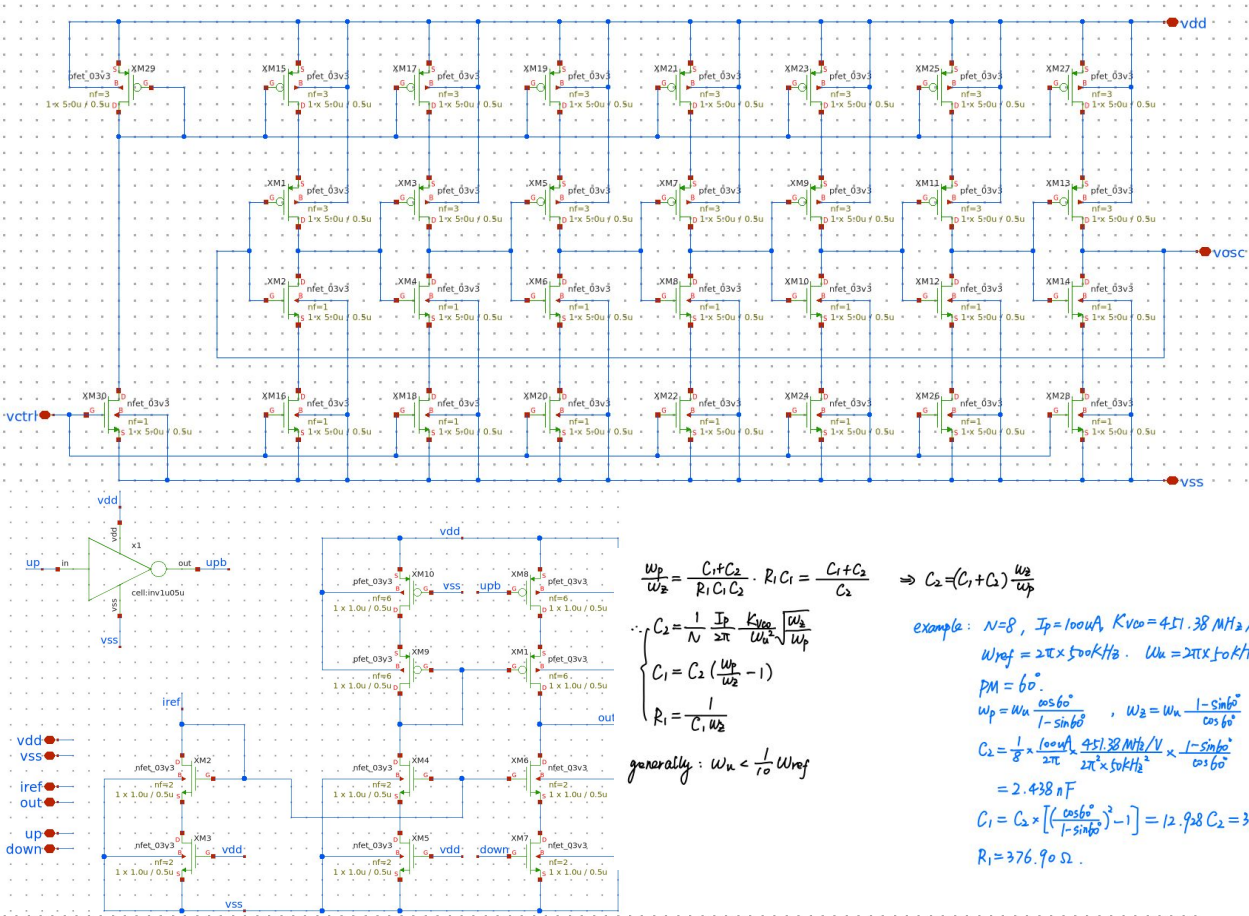
RF_MOSbius

Project information

- Help facilitate lab experiments for students in IC design learning PLL theory.
- Have different PFD and VCO blocks and programmable Dividers and Charge Pumps to allow students to observe PLL operation at different frequencies and learn trade-offs between various PLL configurations and designs.
- Would eventually like to use the PLL MOSbius in a AM and/or FM radio receiver and/or transmitter block which would require a frequency range of around 1MHz to 100MHz.
 - For now, assumption is made that the reference frequency from an off-chip crystal oscillator (put through a reference divider) will be 10kHz.



Proof of Concept: Simple PLL in Xschem



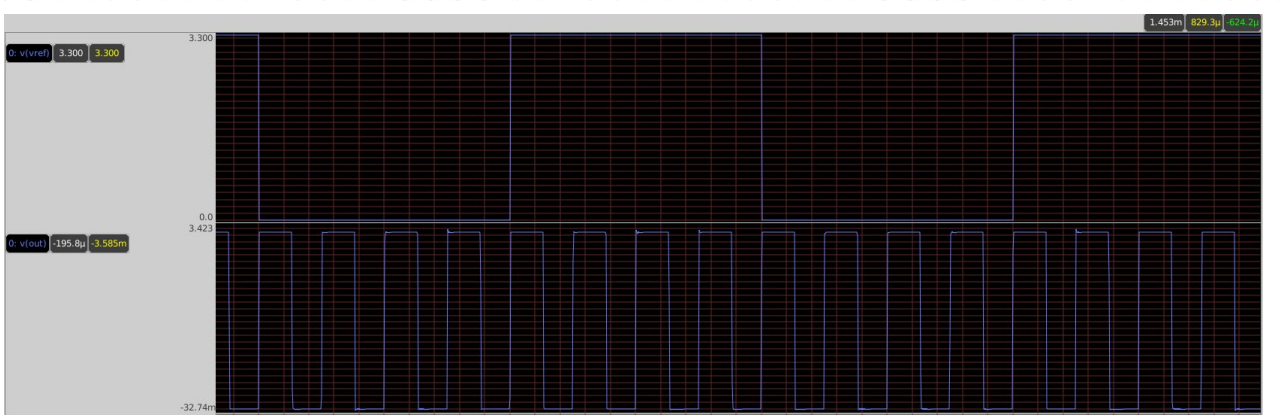
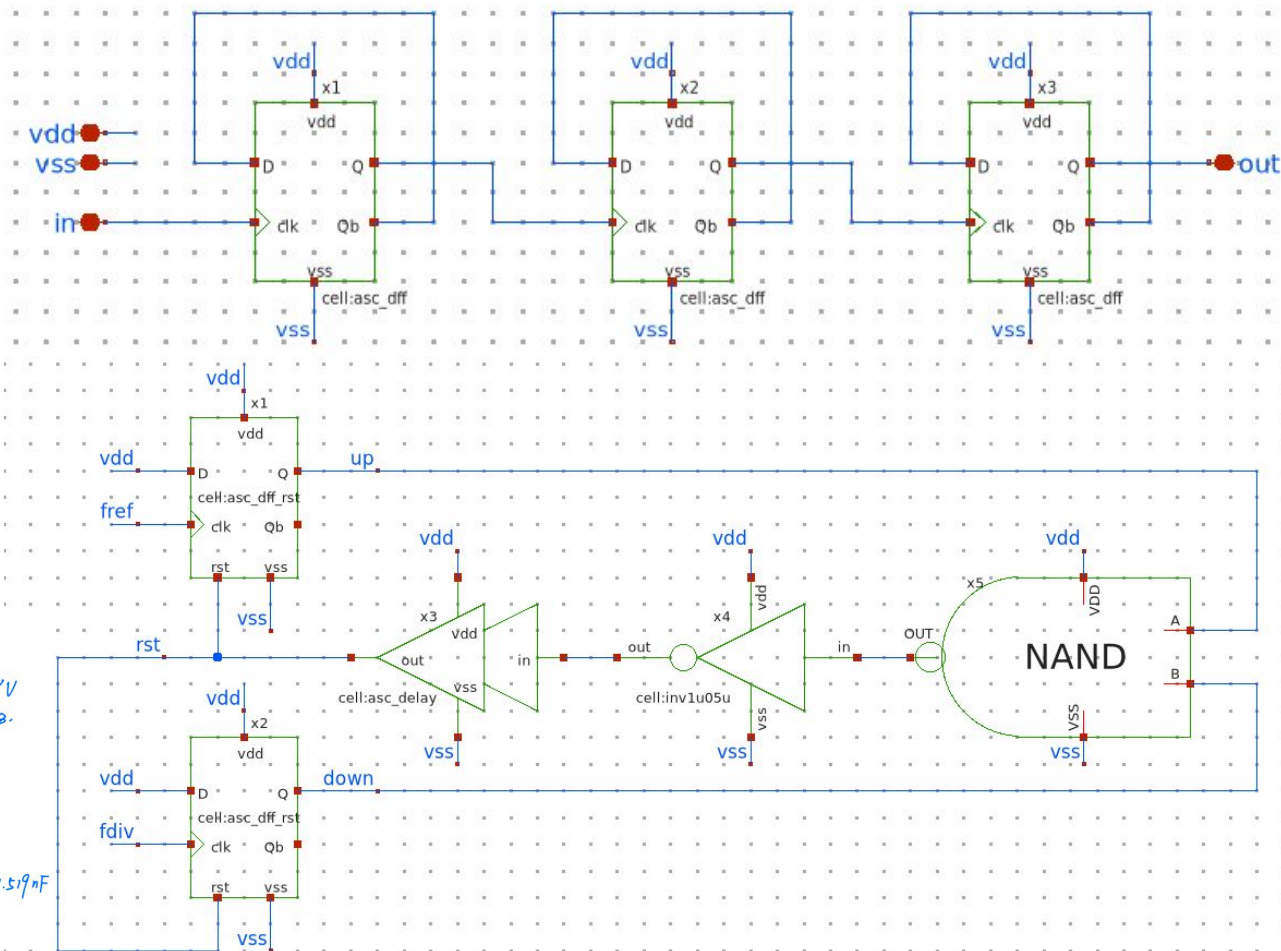
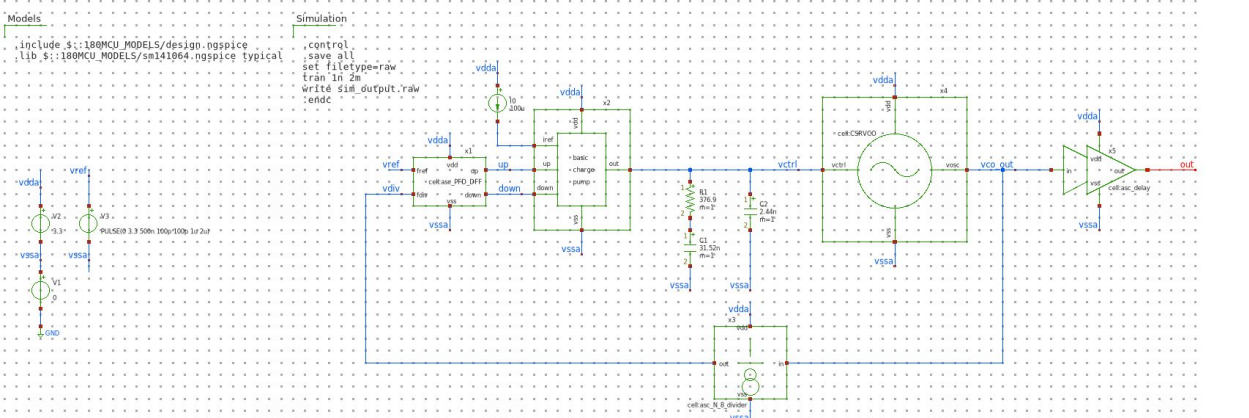
$$\frac{W_p}{W_2} = \frac{C_1 + C_2}{R_1 C_1 C_2} \cdot R_1 C_1 = \frac{C_1 + C_2}{C_2} \Rightarrow C_2 = (C_1 + \frac{W_p}{W_2}) \frac{W_2}{W_p}$$

$$C_2 = \frac{1}{N} \frac{I_p}{2\pi} \frac{K_{VCO}}{W_2} \sqrt{\frac{W_p}{W_2}}$$

$$\left\{ \begin{aligned} C_1 &= C_2 \left(\frac{W_p}{W_2} - 1 \right) \\ R_1 &= \frac{1}{C_1 W_2} \end{aligned} \right.$$

generally: $W_n < \frac{1}{10} W_{ref}$

example: $N=8, I_p=100\mu A, K_{VCO}=451.38 \text{ MHz/V}$
 $W_{ref} = 2\pi \times 500 \text{ kHz}, W_n = 2\pi \times 50 \text{ kHz}$
 $PM = 60^\circ$
 $W_p = W_n \frac{\cos 60^\circ}{1 - \sin 60^\circ}, W_2 = W_n \frac{1 - \sin 60^\circ}{\cos 60^\circ}$
 $C_2 = \frac{1}{8} \times \frac{100\mu A}{2\pi} \times \frac{451.38 \text{ MHz/V}}{2\pi \times 50 \text{ kHz}^2} \times \frac{1 - \sin 60^\circ}{\cos 60^\circ}$
 $= 2.438 \text{ nF}$
 $C_1 = C_2 \times \left[\left(\frac{\cos 60^\circ}{1 - \sin 60^\circ} \right)^2 - 1 \right] = 12.928 C_2 = 31.519 \text{ nF}$
 $R_1 = 376.90 \Omega$



Lead:

Team: Schmith

Project proposal

Lead:

Team: Silicon_BBI2

Project proposal

Lead:

Team: simplymosbius

Project proposal

SpikCore - Spiking mosbius

Team members: Royce Richmond [Team Leader] (RR)
Rosendo Valdés (RV)
Athesh
Venkadesh

Team background:

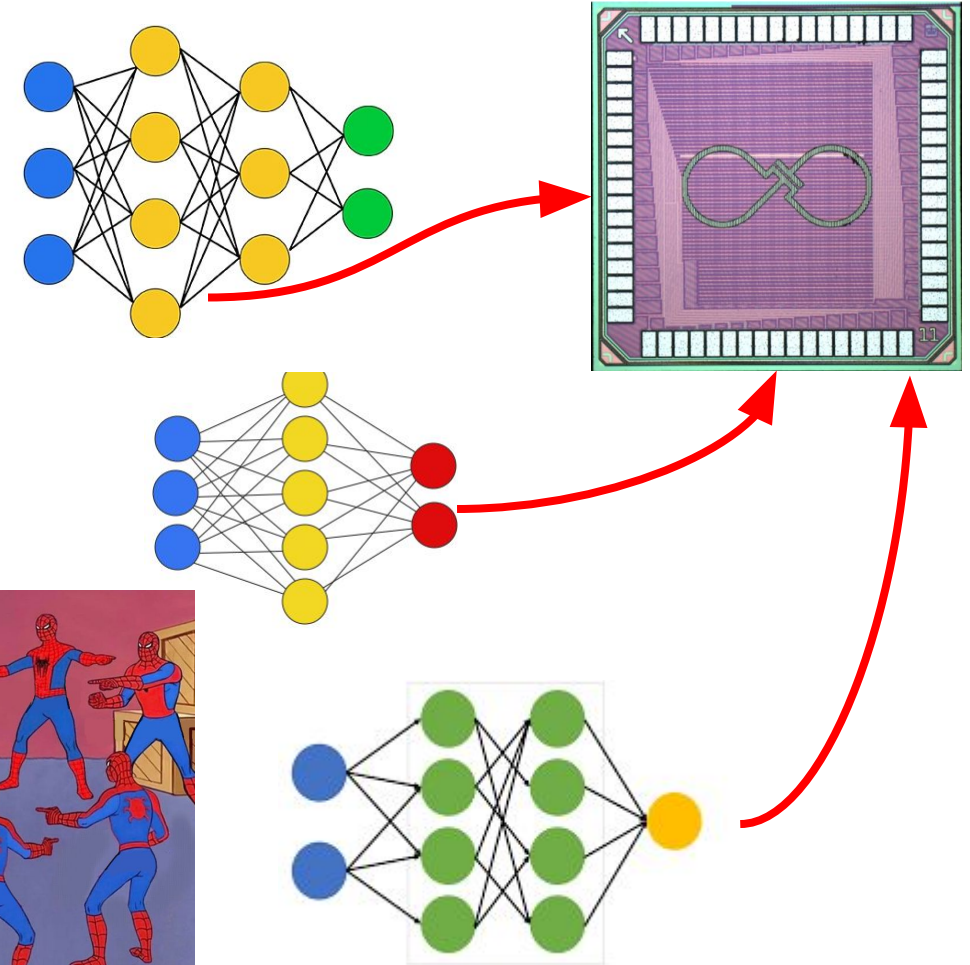
- Academic Experience:
 - RR - Third-year Phd student at CIC - IPN
 - RV - Final-year Master's student at CIC - IPN
- Work Experience:
 - RV - Signal integrity engineer @ Intel

SpikCore - Spiking mosbius

Project goal:

Develop a reconfigurable neuronal architecture leveraging a crossbar array topology, enabling experimentation with different neural networks configurations, input/output patterns, and domain-specific problem-solving.

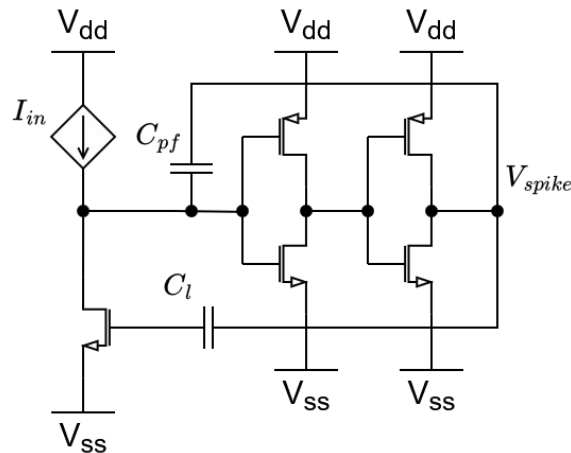
We have
unlimited
power!!
(kind of)



SpikCore - Spiking mosbius

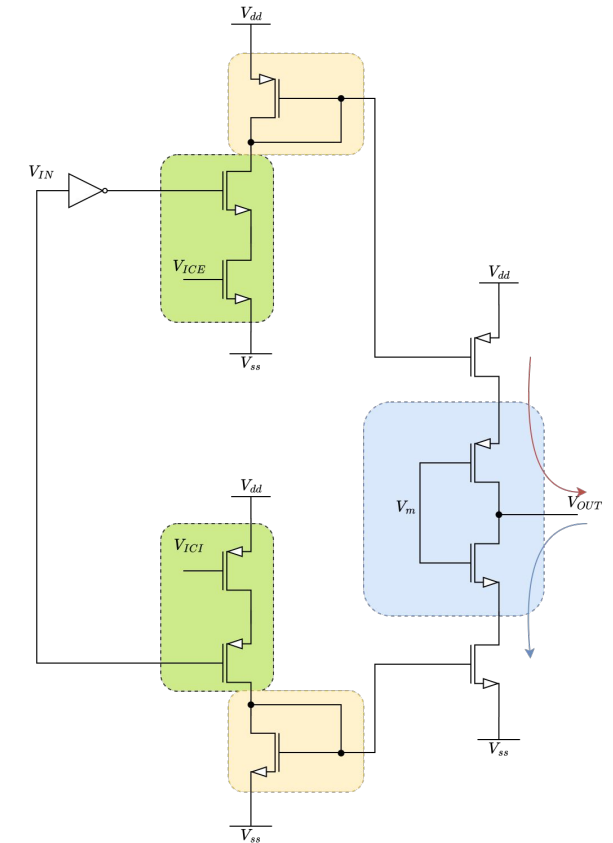
AH (Axon Hillock)[1] neuron

- Small footprint, low power consumption, and high density !!



Synapses

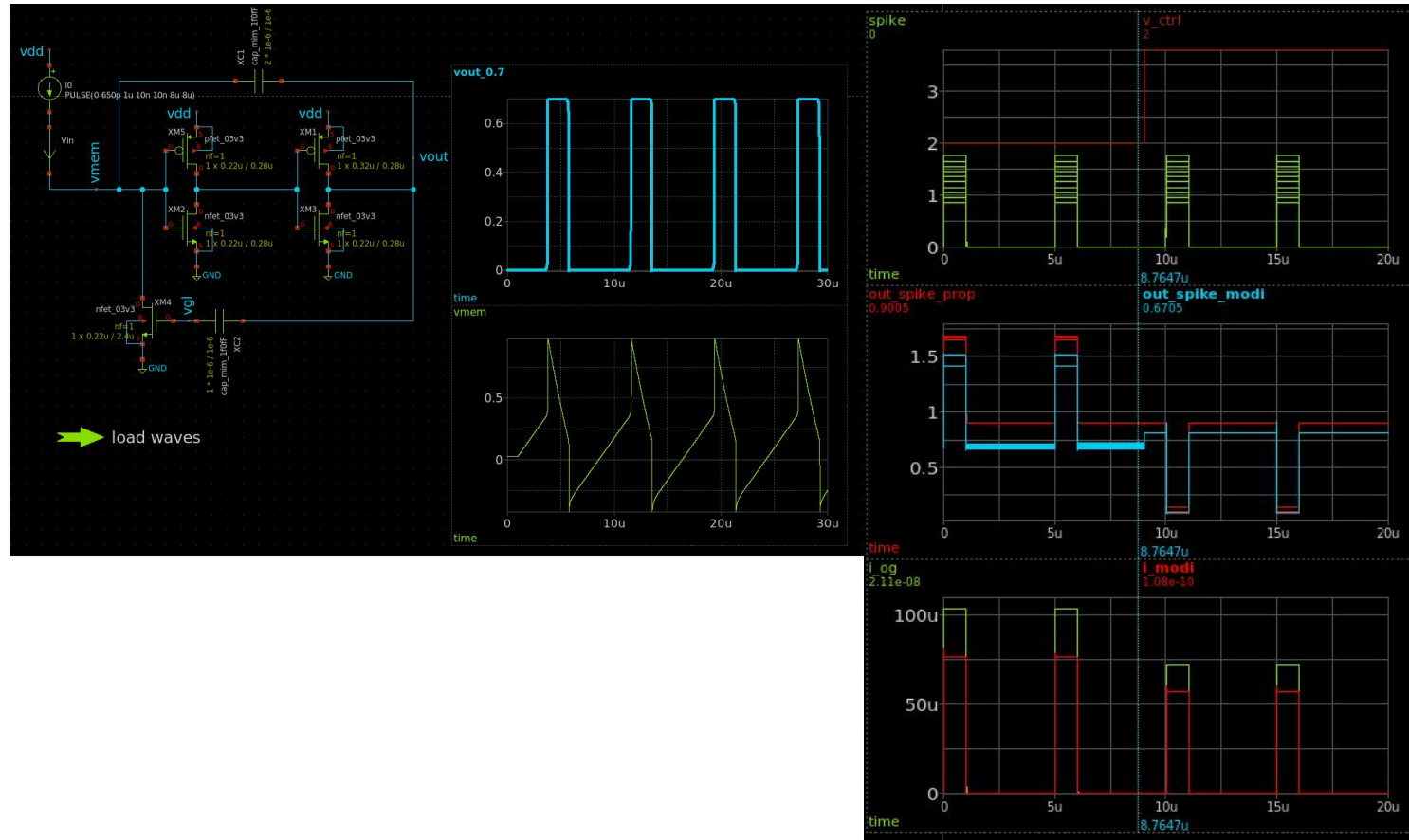
- Pattern recognition relies on excitatory and inhibitory connections, we can replicate this functionality with synapses that can do both.
- Keeping each block simple



[1] Indiveri, G., Linares-Barranco, B., Hamilton, T. J., Schaik, A. van, Etienne-Cummings, R., Delbruck, T., Liu, S.-C., Dudek, P., Häfliger, P., Renaud, S., Schemmel, J., Cauwenberghs, G., Arthur, J., Hynna, K., Folowosele, F., Saighi, S., Serrano-Gotarredona, T., Wijekoon, J., Wang, Y., & Boahen, K. (2011). Neuromorphic Silicon Neuron Circuits. Frontiers in Neuroscience, 5. <https://doi.org/10.3389/fnins.2011.00073>

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Simulation results on neurons, synapses and transmission gates

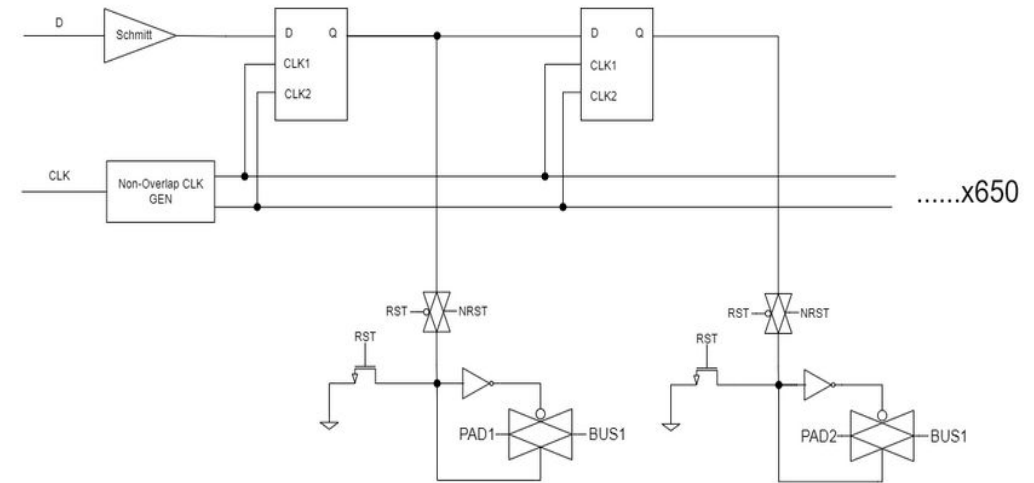


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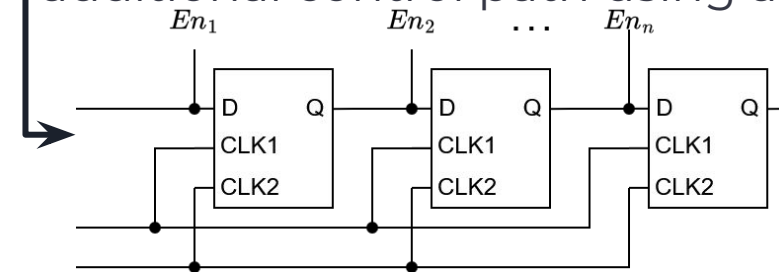
Expand the hardware and software

- expand the flip flops to act as enable signals for subcomponents
- develop (mostly) transparent reconfigurable interconnects between neurons
- extend the bitbang upload method (raspberry pico) to configure the analog architecture

Existing control path for mosbius



additional control path using an overflow

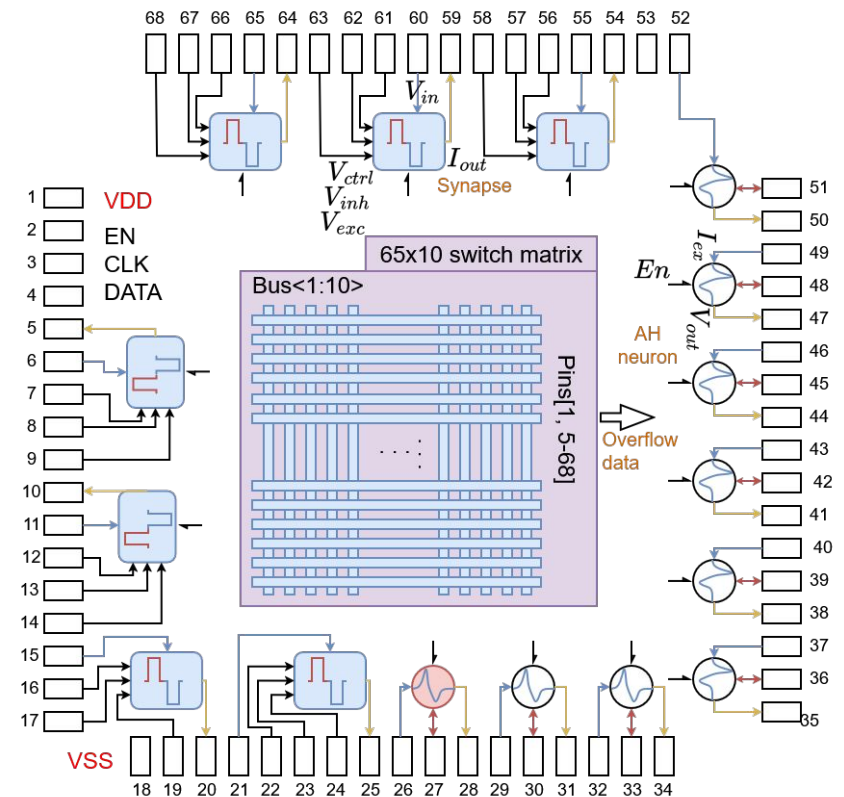
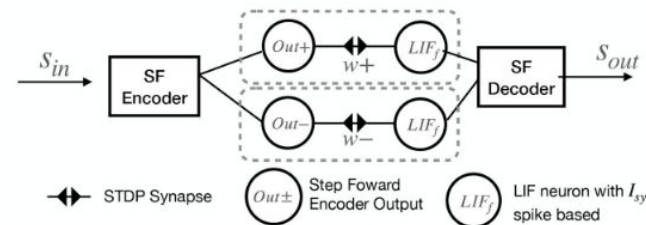


SpikCore - Spiking mosbius

Pinout and characteristics:

- 7 inhibitory/excitatory synapses
- 8 AH neuron subthreshold
- 1 LIF neuron
- 1 LIF neuron subthreshold

The chip can interface with event camera, robotics [2], biophysical signal, etc



SpikCore - Spiking mosbius

Timeline

Week	Task	Responsable	Expected product
28	Project proposal review	All team members	Review proposal with posible updates and tweaks
29	Update and modification to proposal	All team members	Updated project proposal with observations
30	Schematic and symbol creation of neuron, synapse, TG, and read/write subsystem.	Royce Richmond Venkadesh Athesh	Symbol and schematics of building blocks of the peripheral components of spiking mosbius
31	Simulation of TG and characterization	Rosendo Valdés Venkadesh	Simulation and characteristics of TG
32	Simulation of neuron and synapse with TG (pending FF)	Royce Richmond	Simulation and characteristics of neurons and synapses
33	Top cell simulation	All team members	Top cell integration and simulation

SpikCore - Spiking mosbius

Timeline

Week	Task	Responsable	Expected product
34	Layout of individual cells and post layout simulation	Rosendo Valdes Royce richmond	GDS and spice files
35	Layout of Cross-bar array and interconnects	Royce Richmond Venkadesh	GDS and spice files
36	Integration of layouts (top cell) and post layout simulation	All team members	Final GDS file of the top cell
37	Layout review	All team members	Reviewed layout and possible annotations
38	Verification of top cell comparison pre vs post layout	Royce Richmond Rosendo valdes	Final layout ready for submission
39	Review process of submission and repository	Royce Richmond Rosendo valdes	Submission for manufacturing
40	Review of submitted file, simulations and data analysis	Royce Richmond	Report with incidents, improvements and results

Lead:

Team: Tech-8

Project proposal

XDAC

Team members:

Xavier Lee (@xjlee)
Christopher Amankwaa
Devlin Glover
Angel Romero

Team background

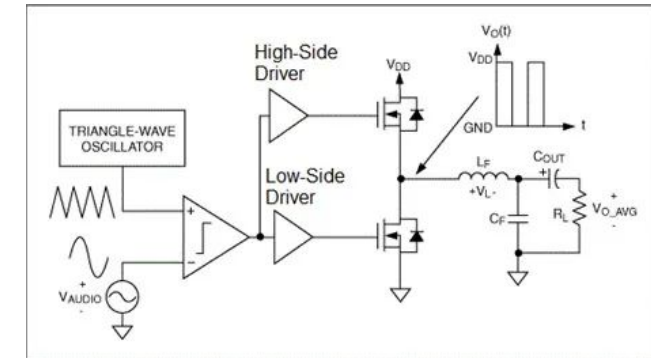
- Academic Experience
2025 graduates of Brown U. (EE/CE) program
Two incoming graduate students in IC design
- Work Experience
...nothing in IC design

XDAC

Project information

- **Goal:** Design a fully operational MOSbius-style configurable Class D amplifier for audio applications with a comparator & push-pull output stage capable of driving typical headphones. The versatility of the individual analog building blocks will permit labs differing from the intended application, our chosen starting point. By including multiple device sizes, different topologies can be used to test, for example, the impacts of transistor sizing or negative feedback.
- **Design:** Necessary fundamental component blocks will include a simple microphone/input amplifier (linear, single-transistor), comparator (possibly with on-board sawtooth or triangle wave oscillator for PWM generation), gate-driver and push-pull power output stage, and the switch matrix/ digital shift register interface to configure all of the amplifier blocks. Current sources and mirrors of varying dimensions will be included, as well as other relevant standard transistor arrangements (still deciding on specific details).
- **Application:** Audio amplifier lab with simple microphone input with options to change transistor sizes.
- **References:**

<https://www.powerelectronicsnews.com/class-d-power-amplifier/>



Questions, Suggestions, Doubts?

Doubts:

Digital aspects: Switch-matrix dimension

Physical Constraints: Land area allotment, packaging?

Questions:

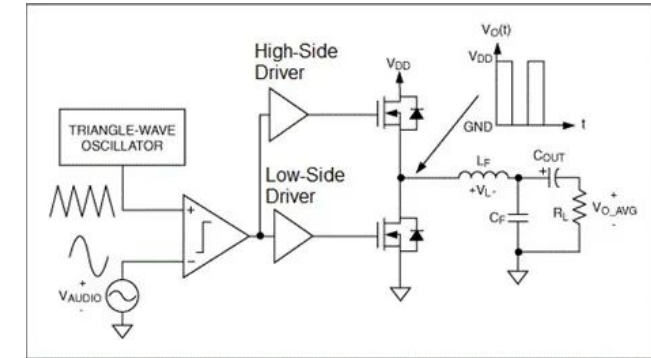
Sawtooth generation on chip? Yay or nay?

Standard cell synthesis from Verilog?

XDAC

Work Divisions

- Xavier: Analog
- Devlin: Digital
- Angel: Digital
- Christopher: Analog



Estimated Transistor Count

Output Drivers: (5+) Two push-pull pairs, plus gate driver circuit

Current Mirrors: (10) Two high-side current mirrors, one low-side. Offer some alternative scaling on select mirrors (1x, 2x, 5x, 10x)

Diff Pairs: (6) Two p-pairs (different sizes), one n-pair.

Basic Op-Amp (8):

Schmitt-Trig Comparator (?):

Bare NMOS: (3)

Bare PMOS: (3)

+ Digital circuitry (unknown quantity)